Artificial Intelligence and the Skill Premium: An Endogenous Growth Perspective

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May 5, 2026

Abstract

This paper investigates the relationship between artificial intelligence and the skill premium from an endogenous growth perspective. It develops a dynamic general equilibrium framework in which AI affects wage inequality through two competing channels: automation, which substitutes capital for labor in existing tasks, and task creation, which expands the range of complex labor-intensive activities. The model further incorporates a learning-by-doing mechanism through which low-skill workers gradually improve their task-specific productivity when exposed to AI-enabled production processes. This mechanism mitigates the initial widening of wage differentials generated by skill-biased technological change and helps explain the bounded evolution of the skill premium observed in advanced economies. The analysis highlights the importance of endogenous labor adaptation in shaping the long-run distributional consequences of AI-driven technological progress.

Keywords: Artificial Intelligence, Skill Premium
JEL Classification: O33, J31, O41

1 Introduction

Artificial intelligence (AI) and automation technologies have emerged as central forces behind a new industrial revolution, fundamentally transforming labor markets. Recent decades have witnessed an exponential increase in the deployment of operative industrial robots and a surge in private investment in AI systems worldwide (Abeliansky et al., 2020; Prettner and Bloom, 2020; Jurkat et al., 2022). Although these technological advances substantially enhance total factor productivity and promote long-run economic growth (Graetz and Michaels, 2018; Kromann et al., 2020), they also generate profound disruptions to income distribution. A central concern in this macroeconomic transition is the impact of AI on the skill premium—that is, the wage differential between high-skill and low-skill workers. Maintaining a stable skill premium is critical for preventing the further exacerbation of income inequality. Accordingly, understanding the causal mechanisms through which endogenous technological change affects the relative demand for heterogeneous skills remains a pivotal challenge in contemporary economic research.

The existing literature on skill-biased technological change and capital-skill complementarity provides foundational frameworks for analyzing wage inequality (Acemoglu, 2002; Krusell et al., 2000; Duffy et al., 2004). Within this literature, automation is frequently modeled as an asymmetric process of task replacement. Empirical and theoretical studies suggest that traditional automation and industrial robots predominantly substitute for low-skill workers performing routine manual tasks. This displacement reduces the relative demand for low-skill labor and thereby places upward pressure on the skill premium (Lankisch et al., 2019; Acemoglu and Restrepo, 2020; Dauth et al., 2021; Cords and Prettner, 2022).

The recent emergence of generative AI and advanced machine learning algorithms, however, introduces a structural paradigm shift. Unlike industrial robots, AI can automate non-routine cognitive tasks that were traditionally performed by high-skill professionals (Acemoglu and Restrepo, 2018a; Autor, 2024). Recent frameworks based on nested constant elasticity of substitution (CES) production functions show that, if AI is more substitutable for high-skill labor than low-skill labor is for high-skill labor, its widespread adoption may exert downward pressure on high-skill wages and thereby reduce the skill premium (Bloom et al., 2025).

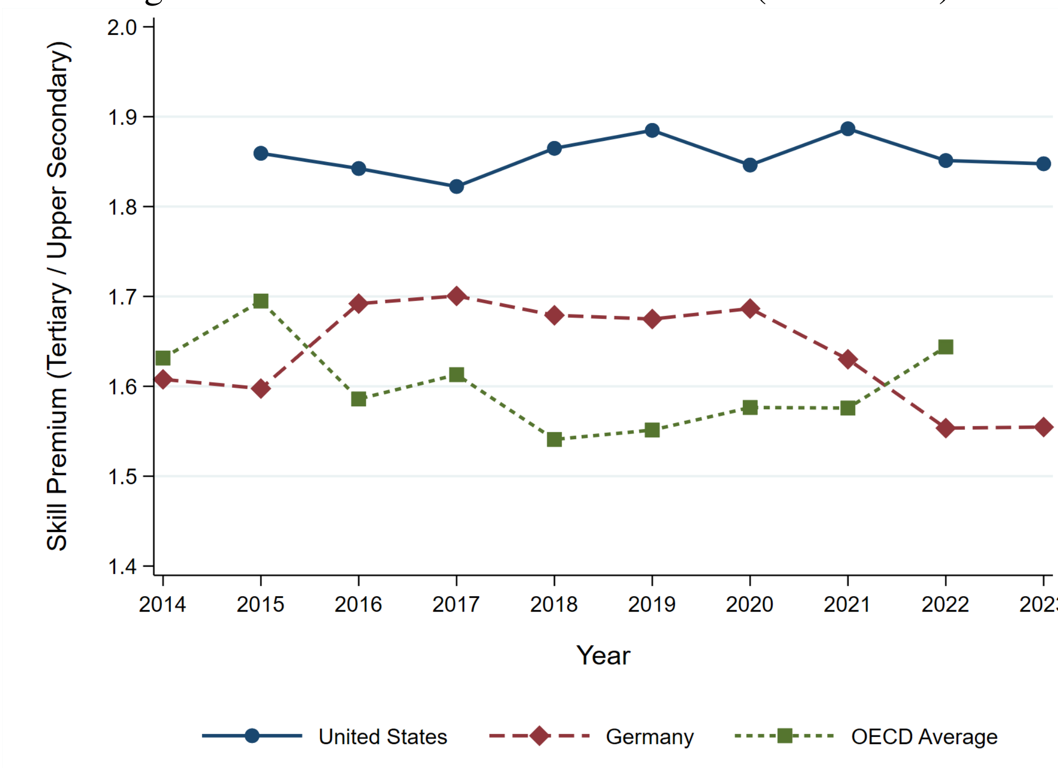
Beyond these static substitution effects, technological progress is inherently dynamic. Following the task-based framework formalized by Acemoglu and Restrepo (2018b), technological change can be decomposed into two distinct types. The first is *automation*, which enables firms to substitute capital for tasks previously performed by labor and generates a displacement effect that typically reduces labor demand. The second is the *creation of new tasks*, whereby old tasks are replaced by new variants in which labor exhibits higher productivity. The reinstatement effect generated by task creation expands labor demand, particularly for high-skill workers who possess an absolute comparative advantage in adapting to new technological paradigms. However, existing task-based models often treat the productivity of heterogeneous labor as temporally fixed following technological shocks, thereby overlooking the endogenous adaptation of the labor force over time.

Imitation, training, and experience accumulation are ubiquitous in human capital development. This dynamic capability is formalized as the "learning-by-doing" effect (Arrow, 1962; Romer, 1986; Stokey, 1988). When confronted with AI-induced task creation, high-skill workers initially capture the technological dividend because of their immediate skill alignment. Over time, however, continuous interaction with new AI interfaces and accumulated production experience allow low-skill workers to endogenously improve their task-specific productivity. This process progressively narrows the efficiency gap between low- and high-skill workers and mitigates initial wage disparities.

2 Empirical Motivation: Stylized Facts

To ground the subsequent theoretical framework in observable macroeconomic dynamics, we construct a harmonized time series of the skill premium across major advanced economies during the recent wave of automation and artificial intelligence from 2014 to 2023. Following the standard labor economics literature, the empirical skill premium (ω_t) is defined as the ratio of median earnings of full-time, full-year workers with tertiary education (ISCED 5–8) to those of workers with upper secondary education (ISCED 3). By restricting the sample to full-time workers, we isolate the structural price of skill, namely the wage rate, from cyclical noise associated with part-time labor supply shocks.

Figure 1: Evolution of the Skill Premium (2014–2023)



Notes: The skill premium (ω_t) is defined as the ratio of median earnings of full-time, full-year workers with tertiary education to those of workers with upper secondary education. The 2014 observation for the United States is omitted because of a structural break in the CPS ASEC reporting methodology for that year.
Source: OECD Education at a Glance Database (2024).

Figure 1 plots the evolution of ω_t for the United States, Germany, and the OECD average. These trajectories reveal two stylized facts that directly inform our structural modeling choices.

Stylized Fact 1: Persistent Structural Divergence. A persistent level gap exists between the United States and European economies. Throughout the observation window, the U.S. skill premium fluctuates around a relatively elevated steady state of approximately 1.85. By contrast, the German skill premium follows a lower and gradually declining trajectory, falling from approximately 1.70 to 1.55, while the broader OECD average remains close to 1.60. This cross-sectional variation suggests that technological shocks, including automation and task creation, do not translate uniformly into wage inequality. Instead, institutional rigidities mediate the transmission of technological change to wages. We hypothesize that stricter European employment protection legislation and more centralized wage bargaining limit the frictionless dismissal of low-skill workers displaced by capital. This institutional constraint forces European firms to internalize the costs of retraining and thereby acts as a catalyst for the learning-by-doing mechanism formalized in Section 3.

Stylized Fact 2: Bounded Trajectories in the Era of AI. Despite the rapid deployment of both industrial robots and generative AI over the past decade (Jurkat et al., 2022), the skill premium has not displayed the explosive divergence predicted by static models of capital-skill complementarity. The trajectories in Figure 1 instead exhibit bounded stability and, in the case of Germany, modest convergence. This empirical pattern is difficult to reconcile with frameworks that assume a perpetual and unilateral bias in technological progress. The data therefore call for a dynamic equilibrium mechanism—specifically, the endogenous learning-by-doing capacity of low-skill labor—that enables displaced workers to gradually transition into moderately complex tasks and structurally anchors the skill premium along a balanced growth path.

3 Model

We develop a three-sector dynamic general equilibrium model to investigate the impact of exogenous AI-driven technological change on the skill premium, explicitly incorporating a learning mechanism. In subsequent sections, we endogenize AI technological progress to provide a deeper understanding of the mechanisms through which AI development shapes the skill premium.

3.1 The Firms' Problem

This subsection characterizes factor demand in the final- and intermediate-goods sectors in the context of AI development. Following Acemoglu and Restrepo (2018b), consider a final-good sector that produces a single homogeneous final output, $Y(t)$, using a continuum of intermediate goods, denoted by $y(i,t)$, distributed over the normalized moving interval $[N(t)-1, N(t)]$. The final good is assembled according to a standard CES production technology:

$$Y(t) = \left(\int_{N(t)-1}^{N(t)} y(i,t)^{\frac{\sigma-1}{\sigma}} \mathrm{d}i \right)^{\frac{\sigma}{\sigma-1}}. \tag{1}$$

All firms in the final- and intermediate-goods sectors operate in perfectly competitive markets. Each intermediate good $y(i,t)$ is supplied by the intermediate-goods sector. We order the continuum of intermediate tasks such that a higher index i corresponds to greater production complexity and higher product quality. The range of intermediate goods used in production is standardized over the interval $[N(t)-1, N(t)]$. An increase in the technological frontier $N(t)$ represents an upgrading in the quality of intermediate goods, which we define as the "creation effect." This specification captures the idea that advances in AI drive quality improvements in intermediate goods: newly created and more complex intermediate tasks systematically replace older, lower-quality tasks.

We next specify the production technology for intermediate goods. Suppose that the production of each intermediate good requires two types of inputs. The first is an AI-driven technology product, denoted by $q(i,t)$. The patent for each $q(i,t)$ is held by a monopolistic firm that can produce it at marginal cost $\mu\psi$, where $\mu \in [0,1]$ and $\psi > 0$. Alternatively, each $q(i,t)$ can be replicated by a competitive firm at the higher marginal cost ψ. Consequently, the equilibrium price of the technology product is strictly tied to ψ. The second type of input consists of capital and labor. We distinguish between low-skill and high-skill labor based on their productivity. For the production of intermediate good i, let $k(i,t)$ denote capital input, while $l(i,t)$ and $h(i,t)$ denote low-skill and high-skill labor inputs, respectively. Furthermore, $\gamma_L(i,t)$ and $\gamma_H(i,t)$ denote the corresponding productivities of low- and high-skill labor.

As AI develops, machines such as operative industrial robots automate many production tasks. However, machines continue to face limitations in replacing human labor in the production of high-quality intermediate goods. To formalize this mechanism, we assume that capital, low-skill labor, and high-skill labor are perfectly substitutable factors of production. We introduce an automation technology threshold, $I(t)$. Labor can produce all intermediate goods, whereas machines can automate production only when $i \leq I(t)$. In this case, firms may choose among all three factors of production. Conversely, when $i > I(t)$, machines are unable to perform the required production tasks, and these intermediate goods can be produced only by labor. Consistent with Acemoglu and Restrepo (2018a), we define an AI-induced increase in the automation frontier $I(t)$ as the "substitution effect." The intermediate-goods production function takes the Cobb–Douglas form:

$$y(i,t) = \begin{cases} q(i,t)^{\eta}\left[k(i,t) + \gamma_L(i,t)l(i,t) + \gamma_H(i,t)h(i,t)\right]^{1-\eta}, & i \leq I(t), \\ q(i,t)^{\eta}\left[\gamma_L(i,t)l(i,t) + \gamma_H(i,t)h(i,t)\right]^{1-\eta}, & i > I(t), \end{cases} \tag{2}$$

where $\eta \in (0,1)$ denotes the factor share of the AI-driven technology product in intermediate-goods production.

Using the production function for intermediate goods $y(i,t)$, we derive the unit cost function $c(\cdot)$ in Appendix A.1. This function is expressed in terms of factor prices. Let $R(t)$, $W_L(t)$, and $W_H(t)$ denote the rental rate of capital, the wage rate of low-skill labor, and the wage rate of high-skill labor, respectively. We define the skill premium as the relative wage of high-skill to low-skill labor, namely $W_H(t)/W_L(t)$. To account for heterogeneity in labor productivity across tasks, we define the effective wage rates of low- and high-skill labor as $W_L(t)/\gamma_L(i,t)$ and $W_H(t)/\gamma_H(i,t)$, respectively. Under perfect competition, the price of each intermediate good $p(i,t)$ is driven down to its minimum unit production cost:

$$p(i,t) = \begin{cases} \min\left\{c(R(t)), c\left(W_L(t)/\gamma_L(i,t)\right), c\left(W_H(t)/\gamma_H(i,t)\right)\right\}, & i \leq I(t), \\ \min\left\{c\left(W_L(t)/\gamma_L(i,t)\right), c\left(W_H(t)/\gamma_H(i,t)\right)\right\}, & i > I(t). \end{cases} \tag{3}$$

The unit cost function is given by

$$c(x(t)) = \eta^{-\eta}(1-\eta)^{\eta-1}\psi^{\eta}x(t)^{1-\eta}. \tag{4}$$

Different factors of production exhibit heterogeneous productivities when producing the same intermediate good. On the one hand, as the quality of an intermediate good increases, labor generally exhibits a productivity advantage over machines. On the other hand, in the production of the most complex intermediate goods, high-skill labor adapts more readily to new technologies and therefore exhibits higher productivity. To formalize this skill-task matching mechanism, which builds on the theories of Costinot and Vogel (2010) and Acemoglu and Autor (2011), we impose the following assumption on labor productivity functions.

Assumption 1 (Skill-task comparative advantage)**.** *The labor productivity functions satisfy*

$$\frac{\partial \gamma_L(i,t)}{\partial i} > 0, \qquad \frac{\partial \gamma_H(i,t)}{\partial i} > 0, \qquad \frac{\partial}{\partial i}\left(\frac{\gamma_H(i,t)}{\gamma_L(i,t)}\right) > 0. \tag{5}$$

Assumption 1 states that both labor productivity functions are strictly increasing in i. By normalizing capital productivity to one, this assumption implies that, as i increases, labor productivity rises while capital productivity remains constant. Labor therefore has a strict comparative advantage over capital in producing higher-quality intermediate goods. Likewise, the ratio $\gamma_H(i,t)/\gamma_L(i,t)$ is strictly increasing in i, implying that high-skill labor has a strict comparative advantage over low-skill labor in producing high-quality intermediate goods.

Given Assumption 1, the expression for $p(i,t)$ implies the existence of a binding production threshold $\tilde{I}(t) = \min\{I(t), I_L(t), I_H(t)\}$. Firms use capital to produce intermediate goods only when $i \le \tilde{I}(t)$. The thresholds $I_L(t)$ and $I_H(t)$ are economic thresholds at which the capital rental rate equals the effective wage rates of low- and high-skill labor, respectively:

$$W_L(t)/\gamma_L(I_L(t)) = R(t), \qquad W_H(t)/\gamma_H(I_H(t)) = R(t). \tag{6}$$

Intermediate-goods producers substitute machines for low-skill labor only when $i < I_L(t)$, meaning that the rental rate of capital is strictly lower than the effective wage of low-skill labor. A symmetric cost-minimization logic governs the trade-off between capital and high-skill labor. Aggregating these structural thresholds yields the binding cutoff index for automated production:

$$\tilde{I}(t) = \min\{I(t), I_L(t), I_H(t)\}. \tag{7}$$

Following Assumption 1, there exists an endogenous skill-task threshold $M(t)$ at which the unit costs of employing low- and high-skill labor are equalized:

$$\frac{W_L(t)}{\gamma_L(M(t))} = \frac{W_H(t)}{\gamma_H(M(t))}. \tag{8}$$

At $i = M(t)$, intermediate-goods firms are indifferent between hiring low- and high-skill labor. For any intermediate good satisfying $i < M(t)$, Assumption 1 ensures that $W_L(t)/\gamma_L(i,t) < W_H(t)/\gamma_H(i,t)$ holds strictly. Consequently, low-skill labor has a comparative advantage and lower effective unit cost in relatively less complex intermediate goods. Conversely, for highly complex intermediate goods with $i > M(t)$, the effective unit cost of high-skill labor is minimized. This factor allocation mechanism is consistent with the skill-biased task framework formalized by Acemoglu and Autor (2011). To isolate the analytical focus on the skill premium, we impose the following boundary condition on automation penetration.

Assumption 2 (Interior automation frontier)**.** *The exogenous automation frontier*

$I(t)$ is sufficiently small relative to the production interval of intermediate goods, satisfying

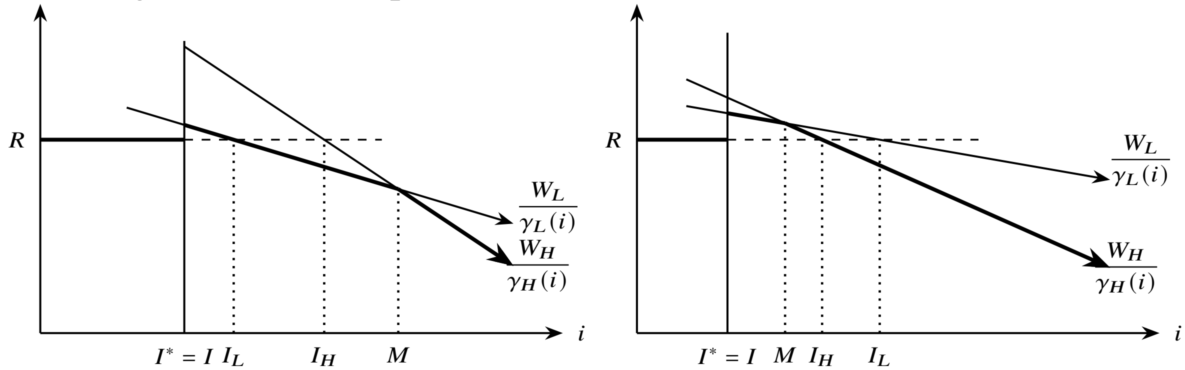
$$I(t)^* = \min\{I(t), I_L(t), I_H(t), M(t)\} = I(t). \tag{9}$$

Assumption 2 formally excludes corner solutions in which capital entirely displaces human labor. As $I(t)$ increases, intermediate-goods firms marginally substitute cheaper capital for labor, a displacement dynamic consistent with the routine-replacement hypothesis proposed by Autor et al. (2003). Without this binding constraint, an expansion in automation capabilities would not alter aggregate factor demand or factor prices. In that case, the skill premium would remain unchanged.

Under Assumptions 1 and 2, the two production thresholds can be illustrated by Figure 2. The bold segments in the figure represent the production factors selected by intermediate-goods producers. The capital rental rate $R(t)$ is represented by a horizontal line because capital productivity is normalized to one and therefore does not vary with the task index i. By contrast, under the monotonicity assumptions imposed on $\gamma_L(i,t)$ and $\gamma_H(i,t)$, the effective wage rates $W_L(t)/\gamma_L(i,t)$ and $W_H(t)/\gamma_H(i,t)$ both decline as i increases. This implies that, as production complexity rises, the unit cost of labor-based production falls because labor has a comparative advantage in more complex tasks. This task-based allocation mechanism is consistent with the assignment and skill-task matching frameworks of Costinot and Vogel (2010) and Acemoglu and Autor (2011).

To parsimoniously formalize the trajectory of AI advancement, we define $n(t) \equiv N(t) - I(t)$ as a measure of labor-intensive tasks, namely the measure of intermediate goods produced by labor. Following the task framework of Acemoglu and Restrepo (2018b), the successful development of job-creating technology leads to the emergence of new and complex intermediate goods, thereby increasing $N(t)$. This expands $n(t)$ because machines remain unable to automate these newly created tasks. Conversely, the successful development of job-replacing technology raises the automation frontier $I(t)$, enabling capital to displace labor in the production of relatively less complex intermediate goods and thereby reducing $n(t)$.

Figure 2: Factor Input Choice in Intermediate-Goods Production



(a) Low-skill labor as the main input (b) High-skill labor as the main input

Notes: The bold segments indicate the factor inputs selected by intermediate-goods producers. In each panel, the bold slanted segment is the lower envelope of the two cost schedules, $\min\{W_L/\gamma_L(i), W_H/\gamma_H(i)\}$. Panel (a) illustrates the case in which low-skill labor is the main labor input, whereas Panel (b) illustrates the case in which high-skill labor is the main labor input. The curves represent monotonic trends rather than exact functional forms.

Furthermore, the allocation of tasks between high- and low-skill labor depends on the endogenous threshold $M(t)$, which is determined by the aggregate AI development state $\{I(t), N(t)\}$. We likewise define $m(t) \equiv M(t) - I(t)$ as the measure of intermediate tasks primarily performed by low-skill labor.

To facilitate stationary equilibrium analysis along the balanced growth path, we normalize the system variables. Let $g(t)$ denote the exogenous growth rate of high-skill labor productivity evaluated at the automation threshold, $\gamma_H(I(t))$. We define normalized aggregate output, capital, and consumption as $y(t) \equiv Y(t)/\gamma_H(I(t))$, $k(t) \equiv K(t)/\gamma_H(I(t))$, and $c(t) \equiv C(t)/\gamma_H(I(t))$, respectively. Correspondingly, the normalized wage rates for low- and high-skill labor are given by $w_L(t) \equiv W_L(t)/\gamma_H(I(t))$ and $w_H(t) \equiv W_H(t)/\gamma_H(I(t))$. By aggregating factor demands across the task continuum and invoking the zero-profit condition, the market-clearing conditions for factor demand and the ideal price index can be expressed as follows. Detailed derivations and the equilibrium price conditions are provided in Appendix A.2.

$$(1-\eta)y(t)(1-n(t))c(R(t))^{1-\sigma}R(t)^{-1} = k^d(t). \tag{10}$$

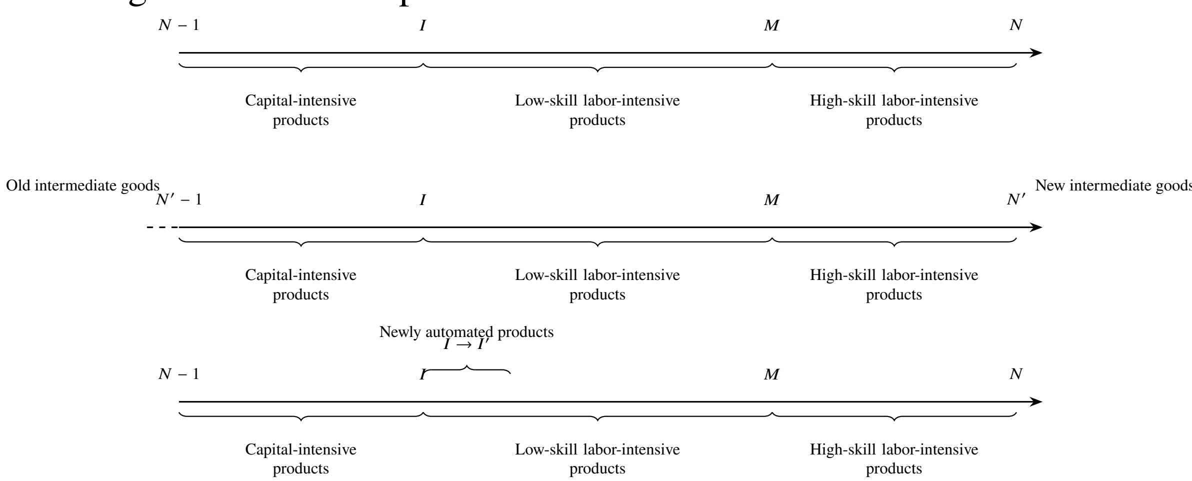
$$(1-\eta)y(t)\int_0^{m(t)} c\left(\frac{w_L(t)}{\gamma_L(i,t)}\right)^{1-\sigma} w_L(t)^{-1}\mathrm{d}i = L^d(t). \tag{11}$$

$$(1-\eta)y(t)\int_{m(t)}^{n(t)} c\left(\frac{w_H(t)}{\gamma_H(i,t)}\right)^{1-\sigma} w_H(t)^{-1}\mathrm{d}i = H^d(t). \tag{12}$$

$$(1-n(t))c(R(t))^{1-\sigma} + \int_0^{m(t)} c\left(\frac{w_L(t)}{\gamma_L(i,t)}\right)^{1-\sigma} \mathrm{d}i + \int_{m(t)}^{n(t)} c\left(\frac{w_H(t)}{\gamma_H(i,t)}\right)^{1-\sigma} \mathrm{d}i = 1. \tag{13}$$

where $k^d(t)$, $L^d(t)$, and $H^d(t)$ denote the demand for capital, low-skill labor, and high-skill labor, respectively. Figure 3 illustrates the equilibrium allocation of factor inputs across different intermediate goods. It also shows how existing intermediate goods are displaced and upgraded when new AI-related technological products are introduced.

Figure 3: Product Space and the Evolution of Intermediate Goods



Notes: This figure illustrates the structure of the product space and its evolution. The interval from $N-1$ to N is divided into capital-intensive products, low-skill labor-intensive products, and high-skill labor-intensive products. The middle panel shows the expansion from old to new intermediate goods, while the bottom panel highlights newly automated products in the interval from I to I'.

3.2 The Household's Problem

This subsection characterizes factor supply in the household sector and introduces the learning mechanism of low-skill labor. In response to the rapid rise of artificial intelligence, low-skill workers must adapt to new technologies and evolving skill requirements by learning from high-skill workers or participating in training programs. Continuous learning and repeated task execution during the production process substantially enhance the productivity of low-skill labor. This endogenous productivity enhancement is defined as the "learning-by-doing" effect (Arrow, 1962). We refer to the combined mechanism of learning by doing and participation in training programs as the learning effect. As new technologies diffuse and production experience accumulates, the labor productivity of low-skill workers gradually increases. In the long run, their productivity may converge toward that of high-skill workers. Building on this reasoning, we formalize the productivity functions for both types of labor as follows.

Assumption 3 (Learning-by-doing and labor productivity)**.** *The productivity functions of high- and low-skill labor are given by*

$$\gamma_H(i,t) = e^{A_H i}, \qquad \gamma_L(i,t) = \Gamma(t) e^{A_L i}, \tag{14}$$

where A_H and A_L are the productivity coefficients of high- and low-skill labor, respectively, satisfying $A_H > A_L$.

The term $\Gamma(t)$ captures the learning effect and measures the accumulation of production experience among low-skill workers. This effect is driven by low-skill workers' technological familiarity, $E(t)$, and their total working time, $l_L(t)$. Total working time is divided into training time, $l_T(t)$, and production time, $l_P(t)$. Formally,

$$\Gamma(t) = \beta E(t) l_T(t)^{\alpha} l_P(t)^{1-\alpha}, \tag{15}$$

subject to

$$l_T(t) + l_P(t) = l_L(t). \tag{16}$$

Hence, the optimal time allocation condition for low-skill labor is

$$l_T(t) = \alpha l_L(t), \qquad l_P(t) = (1-\alpha) l_L(t). \tag{17}$$

In Assumption 3, we deliberately abstract from the learning effect for high-skill labor. The rationale is that high-skill workers possess a superior inherent capacity to assimilate new technologies, enabling them to rapidly master the production of complex intermediate goods and achieve peak production efficiency. As a result, high-skill labor is systematically allocated to more sophisticated tasks, whereas routine and repetitive tasks are assigned to low-skill workers.

Through the learning mechanism, the productivity gap between the two skill groups gradually narrows, which may ultimately reduce the skill premium. However, this convergence does not imply that low-skill labor will strictly surpass high-skill labor in productivity. Accordingly, the restriction $\beta < \bar{\beta}$ ensures that, in both the short run and the long run, high-skill labor retains an absolute productivity advantage over low-skill labor across the entire task continuum.

Adapting to AI development requires low-skill workers to invest substantial time and effort in training, thereby incurring an opportunity cost in terms of foregone earnings from alternative employment. The parameter α captures the structural importance of training or learning from high-skill labor during the skill-upgrading process. A higher α implies a longer training duration and, consequently, higher costs for low-skill workers.

Let ρ denote the subjective discount rate, and let $c_L(t)$ and $a_L(t)$ represent normalized consumption and asset holdings per low-skill worker, respectively. The household problem for low-skill labor is

$$\begin{aligned} \max \quad & \int_0^\infty e^{-\rho t} \left[\ln c_L(t) + \ln(1 - l_L(t))\right] \mathrm{d}t \\ \text{s.t.} \quad & \dot{a}_L(t) = (R(t) - g(t)) a_L(t) + \theta w_L(t) l_T(t) + w_L(t) l_P(t) - c_L(t), \\ & l_T(t) = \alpha l_L(t), \qquad l_P(t) = (1-\alpha) l_L(t). \end{aligned} \tag{18}$$

During the training process, low-skill workers incur direct out-of-pocket expenses, such as tuition and equipment costs. The parameter θ inversely captures these frictional costs associated with skill acquisition. A lower θ indicates a higher monetary cost of training.

Let $c_H(t)$, $a_H(t)$, and $l_H(t)$ represent normalized consumption, asset holdings, and labor supply for high-skill workers, respectively. The household problem for high-skill labor is given by

$$\begin{aligned} \max \quad & \int_0^\infty e^{-\rho t} \left[\ln c_H(t) + \ln(1 - l_H(t))\right] \mathrm{d}t \\ \text{s.t.} \quad & \dot{a}_H(t) = (R(t) - g(t))a_H(t) + w_H(t)l_H(t) - c_H(t) + (1-\theta)w_L(t)l_T(t)\frac{L}{H}. \end{aligned} \tag{19}$$

Finally, we denote the aggregate populations of low- and high-skill workers by L and H, respectively. The share of high-skill labor is defined as $\omega = H/(L + H)$, which characterizes the exogenous skill structure of the economy. Let $k^s(t)$, $L^s(t)$, and $H^s(t)$ denote the aggregate supply functions for capital, low-skill labor, and high-skill labor, respectively. The market-clearing aggregate supply equations are

$$a_L(t)L + a_H(t)H = k^s(t), \qquad l_L(t)L = L^s(t), \qquad l_H(t)H = H^s(t). \tag{20}$$

3.3 Balanced Growth Path

We define a balanced growth path as an equilibrium path along which $Y(t)$, $C(t)$, $K(t)$, $W_L(t)$, and $W_H(t)$ grow at constant rates, while $R(t)$, $l_L(t)$, and $l_H(t)$ remain constant. We then normalize the relevant variables and characterize the equilibrium conditions that must hold in the normalized economy.

The market-clearing conditions for production factors are given by

$$k^d(t) = k^s(t) = k(t), \qquad L^d(t) = L^s(t), \qquad H^d(t) = H^s(t). \tag{21}$$

The skill-premium equation is

$$\frac{w_H(t)}{w_L(t)} = e^{(A_H - A_L)M(t)} \left\{\beta E(t) \left[\alpha l_L(t)\right]^{\alpha} \left[(1-\alpha) l_L(t)\right]^{1-\alpha}\right\} \equiv w. \tag{22}$$

The household consumption and labor-supply decision rules are

$$c_L(t) = (1 - \alpha + \theta\alpha) w_L(t) \left[1 - l_L(t)\right], \qquad c_H(t) = w_H(t) \left[1 - l_H(t)\right], \tag{23}$$

and the Euler equations imply

$$\frac{\dot{c}_L(t)}{c_L(t)} = \frac{\dot{c}_H(t)}{c_H(t)} = R(t) - g(t) - \rho = 0. \tag{24}$$

The aggregate resource constraint is

$$\left[R(t) - g(t)\right] k(t) + w_L(t) l_L(t) L + w_H(t) l_H(t) H = c_L(t) L + c_H(t) H = c(t). \tag{25}$$

To avoid multiplicity of equilibria, we fix the labor supply of high-skill workers at $\bar{l}_H$. The following proposition characterizes the conditions under which the economy admits a balanced growth path under exogenous technological progress, as well as the properties of key variables along that path.

Proposition 1 (Balanced Growth Path under Exogenous Technological Progress)**.** *Suppose that Assumptions 2 and 3 hold. If $\rho < \tilde{\rho}$ and*

$$\frac{H}{L} < \frac{\tilde{\alpha}\beta}{1 - 2\bar{l}_H},$$

then, in the long run, for all $t \geq T$, if and only if the rates of technological progress satisfy

$$\dot{N}(t) = \dot{I}(t) = \dot{M}(t) = \Delta, \tag{26}$$

and the growth rate of low-skill workers' technological familiarity satisfies

$$\frac{\dot{E}(t)}{E(t)} = (A_H - A_L)\Delta, \tag{27}$$

there exists a unique balanced growth path. Along this path, if

$$\lim_{t \to \infty} n(t) = n^*,$$

then $Y(t)$, $C(t)$, $K(t)$, $W_L(t)$, and $W_H(t)$ all grow at the common rate

$$g = A_H \Delta,$$

while the capital rental rate R, the interest rate, low-skill labor supply l_L, and the skill premium w remain constant. See proof in Appendix B.1.

Proposition 1 provides two existence conditions for the balanced growth path. First, the development rates of job-replacing technological products and job-creating technological products must be identical, that is, $\dot{N}(t) = \dot{I}(t) = \Delta$. This condition implies that, along the balanced growth path of the exogenous technological progress model, the growth rate of the task-allocation threshold $M(t)$ between low- and high-skill labor is aligned with the pace of AI development. In this sense, the equilibrium allocation between low- and high-skill labor is determined by the evolution of artificial intelligence.

Second, the growth rate of low-skill workers' familiarity with AI-related technological products must equal $(A_H - A_L)\Delta$. Because total working time remains constant along the balanced growth path, the learning-by-doing effect also grows at the rate $(A_H - A_L)\Delta$. This implies that, in the long run, learning by doing is not only an important mechanism through which low-skill workers catch up with high-skill workers, but also a key force stabilizing the skill premium. This implication is closely related to the broader literature on human capital accumulation and learning-based productivity growth (Arrow, 1962; Ben-Porath, 1967; Lucas, 1988).

3.4 Comparative Statics

We next examine how AI development affects the skill premium at different horizons through comparative statics. Based on the preceding analysis, we obtain the following proposition.

Proposition 2 (Short-Run Comparative Statics)**.** *Suppose that Assumptions 2 and 3 hold. In the short run, given the capital supply K, the technological levels N and I, and the learning effect Γ, both the creation effect of AI development, represented by an increase in N, and the substitution effect, represented by an increase in I, raise the skill premium. Formally,*

$$\frac{\partial w}{\partial N} > 0, \qquad \frac{\partial w}{\partial I} > 0. \tag{28}$$

See proof in Appendix B.2.

Proposition 2 shows that, in the short run, when low-skill workers have not yet accumulated sufficient work experience, the learning-by-doing effect remains weak. At this stage, both the substitution effect and the creation effect reduce the relative demand for low-skill labor and increase the skill premium.

In equilibrium, if I remains unchanged while N increases, then the threshold M also increases. The underlying mechanism is as follows. On the one hand, when AI development creates new technological products and labor-intensive tasks, and the level of automation remains unchanged, both high- and low-skill workers acquire a comparative advantage over machines. Since labor can produce these new tasks at relatively lower production costs, the demand for labor in the production sector increases. On the other hand, because high-skill workers possess a technological advantage and are more easily allocated to complex intermediate-goods production, the cost advantage of low-skill workers is confined mainly to lower-complexity intermediate goods. Low-skill workers therefore cannot replace high-skill workers in the production of complex intermediate goods. As a result, the relative demand for high-skill labor rises, and the task-allocation threshold M between low- and high-skill labor also increases.

Similarly, if N remains unchanged while I increases, then M also rises. AI development improves the level of automation. If no new complex intermediate goods are created, the risk of machine replacement is borne primarily by low-skill workers. This mechanism also pushes the threshold M upward.

Combining this result with the definition of M, the skill-premium equation implies that an increase in M raises $\gamma_H(M)/\gamma_L(M)$ and therefore increases the skill premium w. Under the substitution effect, AI development expands the automation frontier, enabling machines to replace labor in production. The workers most likely to be replaced are low-skill workers, whereas high-skill workers with advanced or specialized skills are difficult to replace fully. These two forces jointly increase the skill premium. Under the creation effect, AI development generates new demand for complex intermediate-goods production. High-skill workers, because of their greater ability to adapt rapidly to new technologies, occupy a relatively advantageous position. Consequently, the creation effect increases the relative demand for high-skill labor and further intensifies income inequality.

From a long-run perspective, however, the learning-by-doing mechanism changes the distributional consequences of AI development. We therefore obtain the following result.

Proposition 3 (Long-Run Comparative Statics)**.** *Suppose that Assumptions 2 and 3 hold. In the long run, the creation effect of AI development, represented by an increase in N, reduces the skill premium, whereas the substitution effect, represented by an increase in I, raises the skill premium. Formally,*

$$\frac{\partial w}{\partial N} < 0, \qquad \frac{\partial w}{\partial I} > 0. \tag{29}$$

See proof in Appendix B.3.

Proposition 3 indicates that, in the long run, AI development does not necessarily increase the skill premium. The reason is that low-skill workers can improve their familiarity with AI-related technologies and their labor productivity through learning by doing. The different effects of AI-induced task creation and automation on the skill premium are therefore a central feature of the model.

The substitution effect produces the same qualitative impact in both the short run and the long run: it raises the skill premium. This is because the workers most vulnerable to machine replacement are predominantly low-skill workers, whereas high-skill workers face a much lower probability of replacement. As a result, demand for low-skill labor declines, low-skill wages fall, and the skill premium increases.

By contrast, the effect of task creation differs sharply between the short run and the long run. In the short run, high-skill workers enjoy a "technological dividend" because of their comparative advantage in producing complex intermediate goods. This stimulates demand for high-skill labor in the intermediate-goods sector and further raises the skill premium. In the long run, however, low-skill workers continuously learn during production and improve their own labor productivity. The comparative advantage of high-skill labor in complex intermediate-goods production is therefore gradually weakened, while the demand for low-skill labor gradually increases. During the process of AI development, the diffusion and adoption of new technological products, together with imitation and learning by low-skill workers, strengthen the learning-by-doing effect. This raises the productivity of low-skill labor, allows low-skill workers to share part of the gains from technological progress, and ultimately lowers the skill premium.

4 Endogenous AI Technological Progress

Building on the model developed in Section 3, this section endogenizes AI-driven technological progress. The reason is that the development rates of job-replacing technological products and job-creating technological products need not be identical. Therefore, the condition imposed in the preceding exogenous-technology framework, under which the two types of technological progress grow at the same rate, is restrictive. We therefore introduce an AI research sector that hires scientists to develop AI-related technological products. This allows us to examine how

AI development affects the skill premium when technological progress is driven endogenously by research incentives.

4.1 The AI Research Sector

The preceding sections analyze AI development as an exogenously given process. In reality, however, AI development is largely driven by a research sector that produces AI-related technological products and consists of multiple R&D firms. These firms hire scientists to conduct research. Once research is successful, a firm obtains a technological product that can be used in production and earns profits through patent protection or by providing technological support and services at lower cost. The ultimate objective of the research sector is to maximize long-run profits.

In the present model, $q(i,t)$ denotes the technological product supplied by the AI research sector to the intermediate-goods sector. These technological products can be divided into two types. The first type is a job-replacing technological product, which enables intermediate goods that were previously produced by labor to be produced through automation. This type of innovation gives rise to job-replacing technological progress and generates the substitution effect. The second type is a job-creating technological product, which creates new and more complex intermediate goods. This type of innovation gives rise to job-creating technological progress and generates the creation effect.

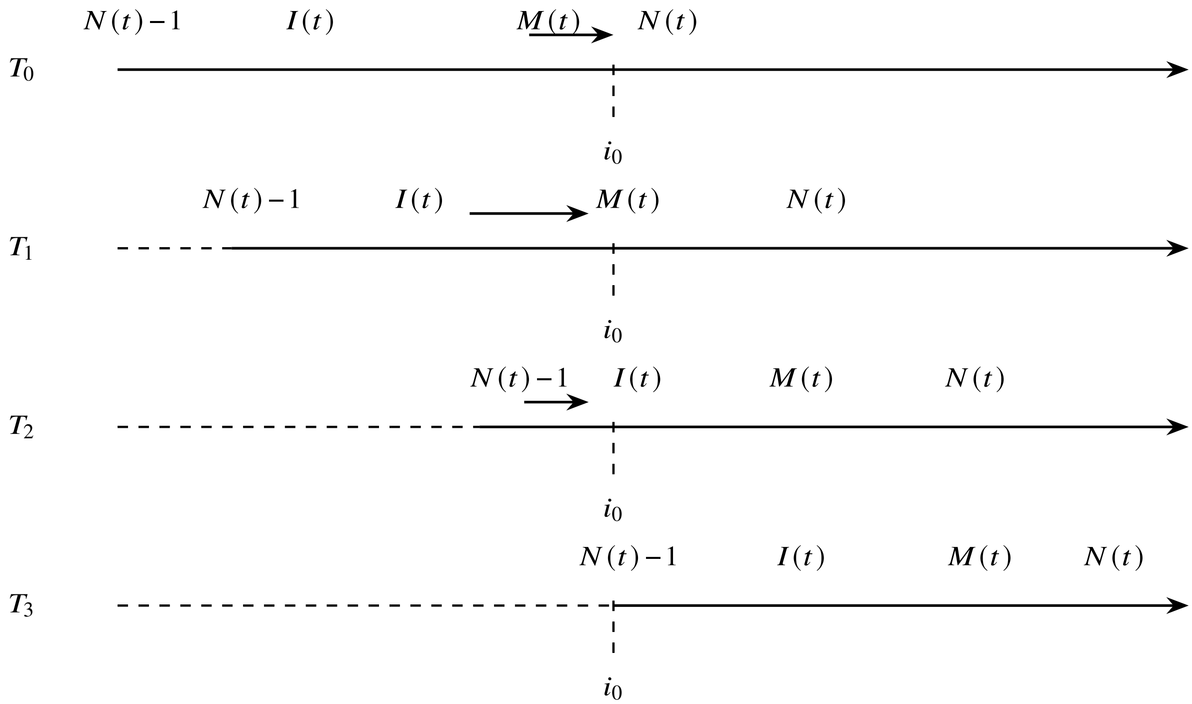
We retain the specification of technological products $q(i,t)$ introduced above. For each $q(i,t)$, there exists an AI research firm that owns the relevant technological patent and can produce the product at marginal cost $\mu\psi$, where $\mu \in [0,1]$ and $\psi > 0$. Competing research firms can produce the same product only at marginal cost ψ. Hence, the equilibrium price of the technological product is ψ, and a patent-holding research firm earns a profit of $(1-\mu)\psi$ for each unit of the technological product it produces. This profit provides the incentive for firms to develop new technological products. This specification follows the standard Schumpeterian view that innovation is motivated by monopoly rents generated by successful research (Aghion and Howitt, 1992; Grossman and Helpman, 1991).

To simplify the analysis, we assume that technological products are always developed sequentially according to the index of intermediate goods, $i \in [0, \infty)$. This implies that, if research on a job-creating technological product is successful, a more complex and refined intermediate good appears, and the frontier $N(t)$ increases. Similarly, if research on a job-replacing technological product is successful, the automation frontier $I(t)$ increases. Following the creative-destruction logic of Schumpeterian growth theory, the continuous development of new technologies renders existing technologies obsolete, thereby eliminating the profits associated with old technological products (Aghion and Howitt, 1992). Once a new job-creating technological product is developed, the lowest-index intermediate good, $N(t) - 1$, is displaced. Once a job-replacing technological product is successfully developed, a labor-intensive product is transformed into a capital-intensive product through automation, and the successful AI research firm obtains the corresponding profit.

Figure 4 illustrates the process through which an intermediate good $i = i_0$ is displaced along the balanced growth path. Because the AI research sector continuously innovates, more complex intermediate goods are developed over time. With the emergence of new tasks, intermediate good i_0 is initially produced by high-skill labor. Through repeated production and learning by low-skill workers, however, i_0 is transformed from a high-skill-labor-intensive product into a low-skill-labor-intensive product at time T_1. As job-replacing technological products are subsequently developed, the automation frontier rises. When automation becomes feasible at time T_2, i_0 is produced by machines and becomes a capital-intensive product. AI continues to develop thereafter, and the intermediate good i_0 is eventually displaced at time T_3.

We assume that the total number of scientists is fixed at S. Scientists can freely choose whether to conduct research on job-replacing technological products or job-creating technological products. Let $S_I(t)$ and $S_N(t)$ denote the numbers of

Figure 4: Dynamic Evolution of Product Ranges Across Periods



Notes: This figure illustrates the dynamic evolution of product ranges over time. The four rows correspond to periods T_0 through T_3. The dashed segments indicate ranges inherited from earlier periods, while the solid segments indicate the contemporaneous product range. The dashed vertical line marks the threshold i_0, which remains fixed across periods and separates the relevant product intervals shown in the figure.

scientists allocated to the two research directions, respectively. They satisfy

$$S_I(t) + S_N(t) \leq S. \tag{30}$$

If a scientist chooses to conduct research on job-replacing technological products, each unit of research time successfully produces κ_I units of technological products, where κ_I denotes the research productivity of job-replacing innovation. The scientist earns the wage $W_I^S(t)$. Similarly, if a scientist chooses to conduct research on job-creating technological products, each unit of research time successfully produces κ_N units of technological products, where κ_N denotes the research productivity of job-creating innovation. The scientist earns the wage $W_N^S(t)$.

Although scientists are free to choose between the two research directions, the cost of conducting research differs across individual scientists because of differences in research specialization, disciplinary background, and individual preferences. For scientist j, the costs of conducting research on job-replacing and job-creating technological products are denoted by $\chi_I^j Y(t)$ and $\chi_N^j Y(t)$, respectively. These costs are scaled by aggregate output because research opportunity costs are assumed to be positively related to the level of economic development. A higher level of aggregate output is associated with a higher cost of time and effort devoted to research.

Scientist j chooses to work on job-replacing technological products if the relative wage gain exceeds the relative cost disadvantage:

$$\frac{W_I^S(t) - W_N^S(t)}{Y(t)} > \chi_I^j - \chi_N^j. \tag{31}$$

To characterize the behavior of scientist allocation near the equilibrium of the research market, we assume that $\chi_I^j - \chi_N^j$ is distributed across scientists according to a cumulative distribution function $G(\cdot)$, where G is continuous and strictly increasing. This implies that, when the wage differential between the two research directions is small, scientists do not switch instantaneously and discontinuously from one research direction to the other. Instead, the allocation of scientists across research activities is smooth.

In addition to scientist allocation, the stage of technological development also affects the relative success rates of the two types of technological progress. The development path of AI-related technological products is characterized by rapid progress in the early stage and slower progress in the later stage, a pattern consistent with the broader literature on technological diffusion and learning curves (Nelson and Phelps, 1966; Griliches, 1957). When the share of labor-intensive production is high, machine-based automation remains relatively underdeveloped. In this case, investment in research is more likely to generate technological breakthroughs in job-replacing products, and the success rate of job-replacing innovation is higher. Conversely, when the automation frontier is already high, further automation becomes more difficult, and the probability of achieving new technological breakthroughs through job-replacing research declines. At the same time, demand for tasks and services that machines cannot provide increases, which raises the success rate of job-creating technological products.

We use the functions $\phi_I(n(t))$ and $\phi_N(n(t))$ to capture these stage-dependent research-success probabilities. Based on the preceding discussion, the development paths of the two types of technological progress are given by

$$\dot{I}(t) = \kappa_I \phi_I(n(t)) S_I(t), \qquad \dot{N}(t) = \kappa_N \phi_N(n(t)) S_N(t). \tag{32}$$

The function $\phi_I(n(t))$ is strictly positive and increasing on $[0, 1]$, whereas $\phi_N(n(t))$ is strictly positive and decreasing on $[0, 1]$. These functions respectively characterize the research-success probabilities of job-replacing and job-creating technological products.

To characterize equilibrium in endogenous technological change, we compute the net present values obtained by the AI research sector from the two types of technological products. Let these values be denoted by $V_I(t)$ and $V_N(t)$. Specifically, $V_I(t)$ represents the value of a successful job-replacing technological product during the stage in which automation has not yet been realized. That is, for $i = I(t) + v$ with $v > 0$, successful job-replacing innovation enables machine-based production and allows the research firm to earn profits through patent protection, ultimately generating the value associated with profit realization. Similarly, $V_N(t)$ represents the value of a successful job-creating technological product. It is the value obtained after profit realization from creating a frontier intermediate good, $i = N(t) + v$.

4.2 Equilibrium with Endogenous Technological Change

We now summarize the economic meanings of the main variables in the endogenous technological progress model. The set $\{I(t), M(t), N(t), K(t)\}$ denotes, respectively, the paths of the automation frontier, the low-skill production threshold, the frontier of intermediate-goods production, and aggregate capital. The set $\{R(t), W_L(t), W_H(t), W_I^S(t), W_N^S(t), Y(t), C(t)\}$ denotes, respectively, the paths of the capital rental rate, the wage rate of low-skill labor, the wage rate of high-skill labor, the wage paid to scientists conducting research on job-replacing technological products, the wage paid to scientists conducting research on job-creating technological products, aggregate output, and aggregate consumption. Finally, the set $\{V_I(t), V_N(t), S_I(t), S_N(t)\}$ denotes, respectively, the paths of the net present values of the two types of technological products and the numbers of scientists allocated to the two research directions.

To characterize the growth paths of these variables in equilibrium, we first compute the profits that AI research firms obtain after successful innovation and then derive the corresponding values after profit realization. For capital-intensive products, the profit from selling the technological product $q(i, t)$ is given by

$$\pi_I(i, t) = (1 - \mu)\eta Y(t) c(R(t))^{1-\sigma}, \qquad N(t) - 1 \leq i \leq I(t). \tag{33}$$

Similarly, for labor-intensive products, using the sign function, the profit generated by the technological product $q(i, t)$ when it is applied to low-skill-labor-intensive and high-skill-labor-intensive production can be written as

$$\pi_N(i,t) = (1-\mu)\eta Y(t)\left[c\left(\frac{W_L(t)}{\gamma_L(i,t)}\right)^{1-\sigma}\frac{\mathrm{sgn}(M(t)-i)+1}{2}\right.$$
$$\left. + c\left(\frac{W_H(t)}{\gamma_H(i,t)}\right)^{1-\sigma}\frac{\mathrm{sgn}(i-M(t))+1}{2}\right], \qquad I(t) < i \le N(t). \tag{34}$$

To express the net present values of the two types of technological products, we define three transition times. Let $T^I(i)$ denote the time required for intermediate good $y(i,t)$ to become automated, so that production shifts from labor to capital. Let $T^M(i)$ denote the time required for low-skill workers to become sufficiently familiar with the production of intermediate good i, so that production shifts from high-skill labor to low-skill labor. Let $T^N(i)$ denote the time required for intermediate good i to be replaced by a more complex frontier intermediate good, after having been produced by high-skill labor.

Along the balanced growth path, AI research firms know the time at which a given technological product will be displaced. Hence, the boundary conditions satisfy

$$V_I(i, T^I(i)) = 0, \qquad V_N(i, T^N(i)) = 0. \tag{35}$$

The value functions are therefore given by

$$V_I(I(t),t) = (1-\mu)\eta\int_t^{T^I(I(t))} e^{-\int_t^\tau [R(s)-\delta]\,\mathrm{d}s}Y(\tau)c(R(\tau))^{1-\sigma}\mathrm{d}\tau, \tag{36}$$

and

$$V_N(N(t),t) = (1-\mu)\eta \int_t^{T^M(N(t))} e^{-\int_t^\tau [R(s)-\delta]\,\mathrm{d}s} Y(\tau) c \left(\frac{W_H(\tau)}{\gamma_H(N(t))}\right)^{1-\sigma} \mathrm{d}\tau$$
$$+ (1-\mu)\eta \int_{T^M(N(t))}^{T^N(N(t))} e^{-\int_t^\tau [R(s)-\delta]\,\mathrm{d}s} Y(\tau) c \left(\frac{W_L(\tau)}{\gamma_L(N(t),\tau)}\right)^{1-\sigma} \mathrm{d}\tau. \tag{37}$$

The derivation of Equations (36) and (37) is provided in Appendix C.1.

Because AI research firms can freely enter the research market, the equilibrium wage paid to scientists conducting research on job-replacing technological products must equal the patent value generated by successful innovation:

$$W_I^S(t) = \kappa_I \phi_I(n(t)) V_I(t). \tag{38}$$

Similarly, the equilibrium wage paid to scientists conducting research on job-creating technological products satisfies

$$W_N^S(t) = \kappa_N \phi_N(n(t)) V_N(t). \tag{39}$$

In addition, scientists trade off research wages against research costs. In equilibrium, research costs equal research wages:

$$W_I^S(t) = \chi_I Y(t), \qquad W_N^S(t) = \chi_N Y(t). \tag{40}$$

It follows that the allocation of scientists across the two research directions satisfies

$$S_I(t) = SG\left(\frac{\kappa_I \phi_I(n(t)) V_I(t)}{Y(t)} - \frac{\kappa_N \phi_N(n(t)) V_N(t)}{Y(t)}\right), \tag{41}$$

and

$$S_N(t) = S\left[1 - G\left(\frac{\kappa_I \phi_I(n(t)) V_I(t)}{Y(t)} - \frac{\kappa_N \phi_N(n(t)) V_N(t)}{Y(t)}\right)\right]. \tag{42}$$

Intuitively, regardless of which type of technological product is being developed, the research direction that yields a higher expected profit enables research firms to offer higher wages and therefore attracts more scientists. Consequently, more scientists are allocated to the technological product with higher excess returns. To simplify notation, define

$$v_I(n) = \lim_{t \to \infty} \frac{V_I(I(t), t)}{Y(t)}, \qquad v_N(n) = \lim_{t \to \infty} \frac{V_N(N(t), t)}{Y(t)}. \tag{43}$$

The derivation of the normalized value functions is provided in Appendix C.2.

Combining the scientist-allocation equations with the laws of motion for the two technological frontiers implies that $\dot{n}(t) > 0$ if and only if

$$\kappa_N \phi_N(n) v_N(n) > \kappa_I \phi_I(n) v_I(n), \tag{44}$$

and the reverse inequality implies $\dot{n}(t) < 0$.

We continue to use the normalized variables and the definition of the balanced growth path introduced above. In the endogenous technological progress model, the endogenous variables must additionally satisfy the following equilibrium conditions. First, the patent values of technological products must satisfy Equations (36) and (37). Second, the allocation of scientists across research directions must satisfy Equations (41) and (42). Third, the growth rates of output $Y(t)$, consumption $C(t)$, capital $K(t)$, the low-skill wage $W_L(t)$, the high-skill wage $W_H(t)$, and high-skill labor productivity at the automation frontier, $\gamma_H(I(t))$, are all equal to

$$g(t) = A_H \kappa_I \phi_I(n(t)) S_I(t). \tag{45}$$

Fourth, because the measure of labor-intensive products is $n(t) = N(t) - I(t)$, we have

$$\dot{n}(t) = \kappa_N \phi_N(n) S - [\kappa_I \phi_I(n) + \kappa_N \phi_N(n)] \, G \left(\kappa_I \phi_I(n) v_I(n) - \kappa_N \phi_N(n) v_N(n)\right) S. \tag{46}$$

We can now state the main equilibrium result for the endogenous technological progress model.

Proposition 4 (Equilibrium with Endogenous Technological Change)**.** *Suppose that Assumption 1′ and Assumption 2 hold. If $\rho < \tilde{\rho}$ and*

$$\frac{H}{L} < \frac{\tilde{\alpha}\beta}{1 - 2\bar{l}_H},$$

and if

$$[\kappa_I \phi_I(n^\alpha) v_I(n^\alpha)]' > [\kappa_N \phi_N(n^\alpha) v_N(n^\alpha)]' \tag{47}$$

holds continuously, then there exists a unique locally asymptotically stable balanced growth path with endogenous technological change. Along this path,

$$N(t) - I(t) = n^\alpha, \tag{48}$$

and

$$\kappa_I \phi_I(n^\alpha) v_I(n^\alpha) = \kappa_N \phi_N(n^\alpha) v_N(n^\alpha). \tag{49}$$

Moreover, $Y(t)$, $C(t)$, $K(t)$, $W_L(t)$, and $W_H(t)$ all grow at the common rate

$$g = \frac{AS\kappa_I \phi_I(n^\alpha) \kappa_N \phi_N(n^\alpha)}{\kappa_I \phi_I(n^\alpha) + \kappa_N \phi_N(n^\alpha)}, \tag{50}$$

the capital rental rate satisfies

$$R = \rho + g, \tag{51}$$

and both low-skill labor supply l_L and the skill premium w remain constant. See proof in Appendix B.4.

Proposition 4 characterizes the equilibrium conditions of the endogenous technological progress model. The variable n^α denotes the equilibrium share of labor-intensive products when AI development is endogenously determined. Because the multi-sector equilibrium system is analytically complex and does not generally yield closed-form solutions for all variables, the next section uses numerical simulations to examine how AI development affects the skill premium and to clarify the underlying mechanisms.

5 Quantitative Analysis

To provide a more intuitive illustration of the preceding theoretical results and to conduct a multi-sector general equilibrium analysis, this section uses numerical simulations to examine how changes in exogenous parameters affect the economy. The equilibrium conditions are jointly characterized by Equations (21) to (25) and Equations (36), (37), (41), (42) and (46).

5.1 Parameter Calibration

The parameters are calibrated using the equilibrium conditions of the model and available macroeconomic, labor-market, and AI-industry data for China. According to the *China Labour Statistical Yearbook* (2021), China's total labor force in 2020 was approximately 770 million. Workers with college education or above accounted for about 22 percent of the labor force. Accordingly, the share of high-skill labor is set to $\omega = 22\%$, while the shares of high- and low-skill labor are set to 22 percent and 78 percent, respectively.

The *China Artificial Intelligence Development Report 2011–2020* reports that China had approximately 20,000 AI-related workers. Based on this proportion, the total labor force in the model is normalized to 100,000, and the number of scientists is set to $S = 2.50$. Following the formula published by the National Bureau of Statistics of China, aggregate labor productivity is measured as the ratio of gross domestic product to total employment. After adjusting employment by the shares of high- and low-skill labor, the productivity parameters of high- and low-skill labor are calibrated as $A_H = 13.30$ and $A_L = 12.03$, respectively.

According to the *China Time Use Survey Bulletin* (2018), the share of total time allocated to work by Chinese residents was 31.5 percent. We therefore set the labor supply of high-skill workers to $l_H = 31.50\%$. In addition, the share of time allocated to training and learning is 1.90 percent. After normalization, this yields $\alpha = 0.06$. According to the *Labor Contract Law of the People's Republic of China*, wages during the probation period cannot be lower than 80 percent of regular wages. We therefore set $\theta = 0.80$.

Following conventional values in the existing macroeconomic literature, the time preference parameter and the elasticity of substitution across intermediate goods are set to $\rho = 0.05$ and $\sigma = 0.90$, respectively. Based on estimates from the China Academy of Information and Communications Technology, the scale of China's core AI industry was approximately RMB 508 billion. Accordingly, the share of AI technological products is set to $\eta = 0.005$. Drawing on the *Artificial Intelligence Development White Paper* (2021) and the *China Artificial Intelligence Patent Technology Analysis Report* (2022), the average annual growth rate of AI patent applications over the previous five years was approximately 32.40 percent. Among these applications, intelligent robots, autonomous driving, and computer vision accounted for about 40 percent. Therefore, the baseline research productivities of job-replacing and job-creating innovation are set to $\kappa_I = 12.96\%$ and $\kappa_N = 19.44\%$, respectively.

The success-rate functions of the two types of technological products are specified as logarithmic functions:

$$\phi_I(n) = \log\left(\frac{1}{n}\right), \qquad \phi_N(n) = \log\left(\frac{1}{1-n}\right). \tag{52}$$

This specification implies that, for a given type of technological product, the probability of research success gradually declines as its application becomes more extensive. It captures the idea that AI development tends to proceed rapidly at an early stage and then slows down as technological opportunities become more difficult to exploit.

The baseline skill premium is measured by the ratio of average wages between high- and low-skill workers. The resulting benchmark value is 2.15. Since the learning-capacity parameter β is difficult to calibrate directly from existing empirical studies and cannot be solved directly from the model, we search over a reasonable parameter range and choose the value that makes the model-generated equilibrium skill premium closest to its empirical counterpart. This procedure yields a baseline value of $\beta = 0.97$. Table 1 reports the benchmark parameter values.

Table 1: Baseline Parameter Values

Parameter	$L + H$	S	ω	A_H	A_L	l_H	β
Value	100000	2.50	22%	13.30	12.03	31.50%	0.97
Parameter	α	θ	ρ	σ	η	κ_I	κ_N
Value	0.06	0.80	0.05	0.90	0.50%	12.96%	19.44%

Notes: The table reports the baseline calibration used in the numerical simulations. The total labor force is normalized to 100,000. The high-skill labor share is denoted by ω, and S denotes the number of scientists in the AI research sector.

5.2 Distributional Effects of AI-Related Parameters

This subsection focuses on how changes in the learning-capacity parameter β, the research productivities of the two types of technological products (κ_I, κ_N), and the labor-force skill structure ω affect the direction and magnitude of changes in the skill premium.

5.2.1 Learning Capacity and the Distributional Effects of AI

We first simulate the effect of changes in the learning capacity of low-skill workers, captured by β. The parameter β reflects a broad set of determinants, including basic education, vocational education, continuing education, and the availability of job-related training for low-skill workers. The simulation results show that, when β increases by 50 percent, the skill premium w falls from 2.15 to 1.49. At the same time, the learning-by-doing effect Γ increases by approximately 42 percent. The share of labor-intensive products n changes only slightly, whereas the share of low-skill-labor-intensive products m rises from 42 percent to 47 percent.

Correspondingly, the share of high-skill-labor-intensive products, $n - m$, falls by about 5 percentage points.

In practice, improvements in the learning capacity of low-skill workers generally reduce the cost of adapting to new technologies and increase the demand for low-skill labor, thereby lowering the skill premium. The literature on education, relative skill supply, and wage inequality provides related evidence. In canonical supply-demand frameworks, an increase in the relative supply of skilled labor tends to reduce the skill premium, holding labor demand constant (Katz and Murphy, 1992; Autor et al., 2008; Goldin and Katz, 2008). More broadly, human capital accumulation and access to education are central mechanisms through which workers adapt to technological change and share in productivity gains.

5.2.2 Research Productivity and the Distributional Effects of AI

We next examine the effects of changes in the research productivity of AI-related technological products. When society continuously introduces automation technologies and increases R&D investment in automation, the research productivity of job-replacing technological products rises. Similarly, when greater investment is devoted to developing complex and refined intermediate goods, or when policies encourage high-quality intermediate-goods innovation, the research productivity of job-creating technological products increases.

When both the substitution-effect coefficient and the creation-effect coefficient are increased by one unit, the simulations show the following results. First, the skill premium rises by 6.48 percent when the substitution effect becomes stronger, whereas it falls by 0.75 percent when the creation effect becomes stronger. This suggests that the research productivity of job-replacing technological products has a more pronounced effect on the skill premium. Second, from the perspective of learning by doing, a stronger substitution effect reduces the demand for low-skill labor and decreases opportunities for task-specific learning, causing the learning-by-doing effect to fall by 6.5 percent. By contrast, the learning-by-doing effect of low-skill workers changes little when the creation effect becomes stronger. Third, in terms of the share of labor-intensive products, a stronger substitution effect reduces this share from 64 percent to 20 percent, whereas a stronger creation effect raises it from 64 percent to 87 percent. Fourth, in terms of low-skill-labor-intensive products, a stronger substitution effect reduces the share from 42 percent to 15 percent, whereas a stronger creation effect increases it from 42 percent to 56 percent. Overall, changes in the research productivity of the two types of technological products have a larger impact on low-skill workers than on high-skill workers.

These results are consistent with the theoretical mechanism developed above. An increase in the research productivity of job-replacing technological products strengthens the substitution effect. AI technological products then become more biased toward automation. Because high-skill workers retain stronger competitiveness in production due to their skill advantage, low-skill workers bear a larger adverse shock. Their wages fall more substantially, which widens income inequality between high- and low-skill workers. By contrast, an increase in the research productivity of job-creating technological products strengthens the creation effect. AI development then generates more complex intermediate goods and creates new tasks. Over time, low-skill workers catch up with high-skill workers through learning by doing and share part of the gains from productivity improvements and technological progress. In the long run, the unit production cost of low-skill labor declines, which prevents further widening of income gaps. Consequently, the skill premium eventually falls as job-creating research productivity increases.

The asymmetric effects of the creation and substitution channels on the skill premium constitute a central feature of the model. Existing studies have not reached a consensus on the distributional consequences of AI. Some evidence suggests that AI differs from earlier automation technologies by being more exposed to high-skill cognitive tasks (Webb, 2020; Autor, 2024). At the same time, the broader automation literature emphasizes that task displacement may reduce labor demand and widen wage inequality when replacement effects dominate (**?**). The present model provides one possible explanation for these mixed implications: the distributional effect of AI depends on whether technological progress is dominated by job-replacing automation or by job-creating task expansion.

5.2.3 Skill Structure and the Distributional Effects of AI

Finally, we examine how changes in the labor-force skill structure affect the distributional consequences of AI. In the baseline calibration, the share of high-skill labor ω is set to 22 percent, corresponding to the share of workers with college education or above in China in 2020. If China continues to expand education and further increases the share of high-skill workers, the wage gap between high- and low-skill workers may be affected.

When the share of high-skill labor rises from 22 percent to 30 percent, the simulations show that the skill premium falls by 7 percent. The learning-by-doing effect increases by 8.6 percent. Changes in the labor-force skill structure have only a limited effect on the share of labor-intensive products, while the share of low-skill-labor-intensive products decreases from 42 percent to 38 percent.

In practice, increasing the share of high-skill workers may reduce the skill premium by changing relative labor supply and weakening the bargaining power of high-skill workers. This result is consistent with the canonical literature on relative labor supply and wage inequality. In the standard supply-demand framework, an expansion in the relative supply of skilled workers lowers the skill premium, conditional on labor demand (Katz and Murphy, 1992; Autor et al., 2008; Goldin and Katz, 2008). In the present model, the AI-related mechanism also works through task reallocation and learning by doing: a larger high-skill workforce weakens the scarcity premium of high-skill labor and facilitates the diffusion of knowledge to low-skill workers through production experience. Consequently, the model predicts a decline in the skill premium as the share of high-skill labor increases.

6 Conclusion

This paper studies how artificial intelligence affects the skill premium in a dynamic general equilibrium framework. The analysis is motivated by the observation that, despite the rapid development of automation and AI technologies, the skill premium in advanced economies has not followed a uniformly explosive path. To rationalize this pattern, the paper develops a task-based endogenous growth model in which AI affects wage inequality through two competing channels: a job-replacing channel that expands the automation frontier and a job-creating channel that generates new and more complex intermediate goods. The model further incorporates a learning-by-doing mechanism through which low-skill workers gradually improve their productivity when they accumulate experience with AI-related production tasks.

The theoretical results show that the distributional effect of AI depends critically on the interaction between task replacement, task creation, and endogenous labor adaptation. In the short run, both automation and task creation tend to raise the skill premium. Automation directly displaces low-skill workers from relatively routine tasks, while task creation initially favors high-skill workers because they are better able to perform newly created complex tasks. In the long run, however, the two effects diverge. The substitution effect continues to increase the skill premium by reducing the demand for low-skill labor. By contrast, the creation effect may reduce the skill premium because new tasks provide low-skill workers with opportunities to accumulate task-specific experience, strengthen their learning-by-doing capacity, and gradually narrow the productivity gap with high-skill workers. Thus, AI development does not have an unambiguous effect on wage inequality. Its long-run distributional consequences depend on whether technological progress is dominated by job-replacing automation or by job-creating task expansion.

The paper also endogenizes AI technological progress by introducing an AI research sector. Scientists choose between developing job-replacing and job-creating technological products, and their allocation depends on the relative expected returns to the two types of innovation. This extension shows that the economy can admit a locally stable balanced growth path with endogenous technological change. Along this path, the relative measure of labor-intensive tasks is determined by an equilibrium arbitrage condition between the value of job-replacing and job-creating innovation. This result highlights that the direction of AI progress is not technologically predetermined. Rather, it is shaped by research incentives, market returns, and the evolving composition of tasks in production.

The quantitative analysis further illustrates the main mechanisms of the model. An increase in the learning capacity of low-skill workers substantially reduces the skill premium by strengthening the learning-by-doing effect and expanding the range of tasks in which low-skill labor can be productively employed. By contrast, an increase in the research productivity of job-replacing innovation raises the skill premium, whereas an increase in the research productivity of job-creating innovation reduces it. Changes in the skill composition of the labor force also matter: a higher share of high-skill workers lowers the skill premium by changing relative labor supply and facilitating the diffusion of knowledge across workers. These results suggest that the distributional consequences of AI are not determined solely by the technical feasibility of automation, but also by workers' adaptive capacity and by the direction of innovation.

The analysis has several policy implications. First, policies that strengthen the learning capacity of low-skill workers can mitigate the inequality-enhancing effects of AI. Investments in basic education, vocational training, continuing education, and workplace-based learning may improve low-skill workers' ability to adapt to new technologies and share in productivity gains. Second, the direction of AI innovation matters. If research incentives are biased toward labor-replacing technologies, AI development is more likely to widen wage inequality. Policies that encourage the creation of new tasks, complementary technologies, and AI applications that augment rather than replace labor may help generate more inclusive growth. Third, the model suggests that the long-run effects of AI depend on the institutional environment that shapes worker retraining, labor mobility, and firm incentives. Institutions that facilitate skill upgrading and task reallocation can play an important role in stabilizing the skill premium.

Several limitations remain. The model abstracts from firm heterogeneity, sectoral differences, and institutional frictions that may affect the diffusion of AI technologies across industries. It also treats high-skill labor as having no explicit learning-by-doing margin, which helps sharpen the mechanism but may understate the adaptive capacity of skilled workers. In addition, the quantitative analysis is illustrative and relies on calibrated parameters rather than structural estimation. Future research could extend the framework by incorporating heterogeneous firms, multiple education groups, endogenous training decisions, and richer forms of AI-human complementarity. Empirical work that separately identifies job-replacing and job-creating AI innovations would also be valuable for testing the model's central prediction: AI raises or lowers the skill premium depending on whether substitution effects or creation effects dominate over time.

Overall, this paper emphasizes that AI should not be viewed as a purely labor-replacing technology. Its impact on wage inequality depends on the dynamic race between automation, task creation, and human learning. When AI primarily substitutes for existing labor tasks, it tends to increase the skill premium. When AI creates new tasks and workers are able to learn and adapt, it can reduce wage inequality in the long run. The key determinant of the distributional outcome is therefore not AI development per se, but the interaction between the direction of technological change and the endogenous adjustment of human capital.

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Appendix A Derivations

A.1 Derivation of the Unit Cost Function

This appendix derives the unit cost function used in the intermediate-goods sector. The derivation corresponds to the production technology in Equation (2) and the unit cost expression in Equation (4). Because the production problem faced by intermediate-goods producers is static, we suppress the time index t throughout this appendix whenever doing so does not cause confusion.

Within the automation frontier, that is, for tasks satisfying $i \leq I$, capital, low-skill labor, and high-skill labor are perfect substitutes in the production of the traditional input bundle. Hence, it is sufficient to derive the unit cost function for one generic input factor. The same argument applies to tasks satisfying $i > I$, where production can be carried out only by labor. We illustrate the derivation using low-skill labor as the input factor. Let

$$x(i) \equiv \frac{W_L}{\gamma_L(i)}$$

denote the effective unit cost of low-skill labor.

When low-skill labor is used in the production of intermediate good i, the production function can be written as

$$y(i) = q(i)^{\eta} \left[\gamma_L(i) l(i)\right]^{1-\eta}. \tag{53}$$

Under perfect competition, cost minimization requires that the marginal rate of technical substitution between the technological product $q(i)$ and low-skill labor equals their relative factor price. Since the price of the technological product supplied by the AI research sector is ψ, and the wage rate of low-skill labor is W_L, the first-order condition is

$$\frac{\psi}{W_L} = \frac{\eta q(i)^{\eta-1} \left[\gamma_L(i) l(i)\right]^{1-\eta}}{(1-\eta) q(i)^{\eta} \left[\gamma_L(i) l(i)\right]^{-\eta} \gamma_L(i)}. \tag{54}$$

Rearranging Equation (54) yields

$$q(i) = \frac{\eta}{1-\eta} \frac{W_L/\gamma_L(i)}{\psi} \gamma_L(i) l(i). \tag{55}$$

Substituting Equation (55) into Equation (53) gives

$$y(i) = \left(\frac{W_L/\gamma_L(i)}{\psi}\right)^{\eta} \left(\frac{\eta}{1-\eta}\right)^{\eta} \gamma_L(i) l(i). \tag{56}$$

Therefore, the conditional demand for low-skill labor is

$$l(i) = \left(\frac{W_L/\gamma_L(i)}{\psi}\right)^{-\eta} \left(\frac{\eta}{1-\eta}\right)^{-\eta} \gamma_L(i)^{-1} y(i), \tag{57}$$

and the corresponding demand for the technological product is

$$q(i) = \left(\frac{W_L/\gamma_L(i)}{\psi}\right)^{1-\eta} \left(\frac{\eta}{1-\eta}\right)^{1-\eta} y(i). \tag{58}$$

The total cost of producing intermediate good i using low-skill labor is therefore

$$\begin{aligned} C(i) &= \psi q(i) + W_L l(i) \\ &= \left(\frac{W_L}{\gamma_L(i)}\right)^{1-\eta} \psi^\eta \left[\left(\frac{\eta}{1-\eta}\right)^{-\eta} + \left(\frac{\eta}{1-\eta}\right)^{1-\eta}\right] y(i) \\ &= \eta^{-\eta}(1-\eta)^{\eta-1}\psi^\eta \left(\frac{W_L}{\gamma_L(i)}\right)^{1-\eta} y(i). \end{aligned} \tag{59}$$

It follows that the unit cost of producing intermediate good i using low-skill labor is

$$c\left(\frac{W_L}{\gamma_L(i)}\right) = \eta^{-\eta}(1-\eta)^{\eta-1}\psi^\eta \left(\frac{W_L}{\gamma_L(i)}\right)^{1-\eta}. \tag{60}$$

The same derivation applies to capital and high-skill labor. Since capital productivity is normalized to one, the unit cost of using capital is

$$c(R) = \eta^{-\eta}(1-\eta)^{\eta-1}\psi^\eta R^{1-\eta}. \tag{61}$$

Similarly, the unit cost of using high-skill labor is

$$c\left(\frac{W_H}{\gamma_H(i)}\right) = \eta^{-\eta}(1-\eta)^{\eta-1}\psi^\eta \left(\frac{W_H}{\gamma_H(i)}\right)^{1-\eta}. \tag{62}$$

More generally, for any input factor with effective factor price x, the unit cost function is

$$c(x) = \eta^{-\eta}(1-\eta)^{\eta-1}\psi^\eta x^{1-\eta}. \tag{63}$$

This expression is identical to the unit cost function in Equation (4).

Because traditional production inputs are perfectly substitutable within the automation frontier, intermediate-goods producers choose the factor with the lowest effective unit cost. Therefore, for $i \leq I$, the unit cost is

$$c\left(\min\left\{R, \frac{W_L}{\gamma_L(i)}, \frac{W_H}{\gamma_H(i)}\right\}\right). \tag{64}$$

For $i > I$, machines cannot perform the required production tasks, so producers choose between low-skill and high-skill labor only:

$$c\left(\min\left\{\frac{W_L}{\gamma_L(i)}, \frac{W_H}{\gamma_H(i)}\right\}\right). \tag{65}$$

Since $c(\cdot)$ is strictly increasing in the effective factor price, the above formulation is equivalent to the intermediate-goods pricing rule in the main text:

$$p(i) = \begin{cases} \min\left\{c(R), c\left(W_L/\gamma_L(i)\right), c\left(W_H/\gamma_H(i)\right)\right\}, & i \leq I, \\ \min\left\{c\left(W_L/\gamma_L(i)\right), c\left(W_H/\gamma_H(i)\right)\right\}, & i > I. \end{cases} \tag{66}$$

Thus, minimizing the effective factor price is equivalent to minimizing the unit production cost. This establishes the unit cost function and the corresponding intermediate-goods price condition used in the main text.

A.2 Derivation of the Equilibrium Price Condition

This appendix derives the equilibrium price condition and the aggregate factor-demand equations used in the main text. The derivation corresponds to the final-good production technology in Equation (1), the intermediate-goods pricing rule in Equation (3), and the normalized market-clearing conditions in Equations (10) to (13). Because the production problem is static, we suppress the time index whenever doing so does not cause confusion.

The representative final-good producer takes the prices of intermediate goods, $p(i,t)$, as given and chooses the demand for each intermediate good $y(i,t)$ to maximize profits:

$$\max_{\{y(i,t)\}} \left[\left(\int_{N(t)-1}^{N(t)} y(i,t)^{\frac{\sigma-1}{\sigma}} \mathrm{d}i \right)^{\frac{\sigma}{\sigma-1}} - \int_{N(t)-1}^{N(t)} p(i,t) y(i,t) \mathrm{d}i \right]. \tag{67}$$

The first-order condition with respect to $y(i,t)$ is

$$Y(t)^{\frac{1}{\sigma}} y(i,t)^{-\frac{1}{\sigma}} = p(i,t). \tag{68}$$

Rearranging Equation (68) yields the demand function for intermediate good i:

$$y(i,t) = p(i,t)^{-\sigma} Y(t). \tag{69}$$

Substituting Equation (69) into the final-good production function gives

$$\begin{aligned} Y(t) &= \left(\int_{N(t)-1}^{N(t)} [p(i,t)^{-\sigma} Y(t)]^{\frac{\sigma-1}{\sigma}} \mathrm{d}i \right)^{\frac{\sigma}{\sigma-1}} \\ &= Y(t) \left(\int_{N(t)-1}^{N(t)} p(i,t)^{1-\sigma} \mathrm{d}i \right)^{\frac{\sigma}{\sigma-1}}. \end{aligned} \tag{70}$$

Hence, the ideal price-index condition is

$$\int_{N(t)-1}^{N(t)} p(i,t)^{1-\sigma} \mathrm{d}i = 1. \tag{71}$$

After the input choice for each intermediate good is determined, the demand for each factor can be obtained by combining Equation (69) with the conditional factor demands derived from the unit cost minimization problem in Section A.1. In particular, if capital is used in the production of intermediate good i, its conditional demand is

$$k^d(i,t) = (1-\eta) c(R(t))^{1-\sigma} R(t)^{-1} Y(t), \qquad i \leq I(t). \tag{72}$$

If capital cannot be used for task i, then capital demand is zero. Therefore,

$$k^d(i,t) = \begin{cases} (1-\eta)Y(t)c(R(t))^{1-\sigma}R(t)^{-1}, & i \le I(t), \\ 0, & i > I(t). \end{cases} \tag{73}$$

Similarly, if low-skill labor is the cost-minimizing input for task i, its conditional demand is

$$l^d(i,t) = (1-\eta)Y(t)c\left(\frac{W_L(t)}{\gamma_L(i,t)}\right)^{1-\sigma} W_L(t)^{-1}. \tag{74}$$

Given the task-allocation threshold $M(t)$, low-skill labor is used for tasks between the automation frontier and the skill threshold. Hence,

$$l^d(i,t) = \begin{cases} 0, & i \le I(t), \\ (1-\eta)Y(t)c\left(\frac{W_L(t)}{\gamma_L(i,t)}\right)^{1-\sigma} W_L(t)^{-1}, & I(t) < i \le M(t), \\ 0, & i > M(t). \end{cases} \tag{75}$$

Analogously, high-skill labor is used for tasks above the skill threshold $M(t)$. Its conditional demand is

$$h^d(i,t) = \begin{cases} 0, & i \le M(t), \\ (1-\eta)Y(t)c\left(\frac{W_H(t)}{\gamma_H(i,t)}\right)^{1-\sigma} W_H(t)^{-1}, & i > M(t). \end{cases} \tag{76}$$

The corresponding aggregate factor demands are obtained by integrating over the relevant task intervals. Since the measure of automated tasks is $1 - n(t)$, aggregate capital demand is

$$K^d(t) = (1-\eta)Y(t)(1-n(t))c(R(t))^{1-\sigma}R(t)^{-1}. \tag{77}$$

Aggregate low-skill labor demand is

$$L^d(t) = (1-\eta)Y(t) \int_{I(t)}^{M(t)} c\left(\frac{W_L(t)}{\gamma_L(i,t)}\right)^{1-\sigma} W_L(t)^{-1}\mathrm{d}i. \tag{78}$$

Aggregate high-skill labor demand is

$$H^d(t) = (1-\eta)Y(t) \int_{M(t)}^{N(t)} c\left(\frac{W_H(t)}{\gamma_H(i,t)}\right)^{1-\sigma} W_H(t)^{-1}\mathrm{d}i. \tag{79}$$

The same input allocation also determines the equilibrium price index. Combining the intermediate-goods pricing rule with Equation (71) yields

$$\begin{aligned}(1-n(t))c(R(t))^{1-\sigma} + \int_{I(t)}^{M(t)} c\left(\frac{W_L(t)}{\gamma_L(i,t)}\right)^{1-\sigma} \mathrm{d}i \\ + \int_{M(t)}^{N(t)} c\left(\frac{W_H(t)}{\gamma_H(i,t)}\right)^{1-\sigma} \mathrm{d}i = 1.\end{aligned} \tag{80}$$

We now transform these conditions into the normalized variables used in the main text. Recall that

$$y(t) \equiv \frac{Y(t)}{\gamma_H(I(t))}, \qquad k(t) \equiv \frac{K(t)}{\gamma_H(I(t))}, \qquad w_L(t) \equiv \frac{W_L(t)}{\gamma_H(I(t))}, \qquad w_H(t) \equiv \frac{W_H(t)}{\gamma_H(I(t))}.$$

Let the transformed task index be $j = i - I(t)$. Then the labor-intensive task interval $[I(t), N(t)]$ is mapped into $[0, n(t)]$, and the low-skill task interval $[I(t), M(t)]$ is mapped into $[0, m(t)]$, where

$$n(t) = N(t) - I(t), \qquad m(t) = M(t) - I(t).$$

For notational simplicity, we relabel the transformed index j as i.

Consider the high-skill labor demand equation as an example. Starting from

Equation (79), we obtain

$$(1-\eta)Y(t)\int_{M(t)}^{N(t)} c\left(\frac{W_H(t)}{\gamma_H(i,t)}\right)^{1-\sigma} W_H(t)^{-1}\mathrm{d}i = H^d(t). \tag{81}$$

Applying the change of variables $i = j + I(t)$ and dividing both sides by the appropriate normalization factor gives

$$(1-\eta)y(t)\int_{m(t)}^{n(t)} c\left(\frac{w_H(t)}{\gamma_H(i,t)}\right)^{1-\sigma} w_H(t)^{-1}\mathrm{d}i = H^d(t). \tag{82}$$

The transformations for capital and low-skill labor are analogous. Therefore, the normalized factor-demand equations are

$$(1-\eta)y(t)(1-n(t))c(R(t))^{1-\sigma}R(t)^{-1} = k^d(t), \tag{83}$$

$$(1-\eta)y(t)\int_{0}^{m(t)} c\left(\frac{w_L(t)}{\gamma_L(i,t)}\right)^{1-\sigma} w_L(t)^{-1}\mathrm{d}i = L^d(t), \tag{84}$$

and

$$(1-\eta)y(t)\int_{m(t)}^{n(t)} c\left(\frac{w_H(t)}{\gamma_H(i,t)}\right)^{1-\sigma} w_H(t)^{-1}\mathrm{d}i = H^d(t). \tag{85}$$

Similarly, the normalized ideal price-index condition is

$$\begin{aligned}(1-n(t))c(R(t))^{1-\sigma} + \int_{0}^{m(t)} c\left(\frac{w_L(t)}{\gamma_L(i,t)}\right)^{1-\sigma} \mathrm{d}i \\ + \int_{m(t)}^{n(t)} c\left(\frac{w_H(t)}{\gamma_H(i,t)}\right)^{1-\sigma} \mathrm{d}i = 1.\end{aligned} \tag{86}$$

Equations Equations (83) to (86) correspond exactly to the normalized equilibrium conditions stated in Equations (10) to (13). This completes the derivation of the equilibrium price condition and the associated market-clearing factor-demand equations.

Appendix B Proofs of Propositions

B.1 Proof of Proposition 1

This proof corresponds to Proposition 1 in the main text.

Suppose that

$$\dot{N}(t) = \dot{I}(t) = \Delta, \qquad \lim_{t\to\infty} n(t) = n^*.$$

Along the balanced growth path, the normalized equilibrium conditions can be written as

$$(1-\eta)y(1-n)c(R)^{1-\sigma}R^{-1} = k, \tag{87}$$

$$(1-\eta)y\int_0^m c\left(\frac{w_L\gamma_H(I)}{\gamma_L(i+I)}\right)^{1-\sigma} w_L^{-1}\mathrm{d}i = l_L L, \tag{88}$$

$$(1-\eta)y\int_m^n c\left(\frac{w_H}{\gamma_H(i)}\right)^{1-\sigma} w_H^{-1}\mathrm{d}i = l_H H, \tag{89}$$

$$(1-\eta)y = c + gk, \tag{90}$$

$$\frac{\dot{E}}{E} = (A_H - A_L)\Delta, \tag{91}$$

$$\tilde{\alpha}\beta l_L w_H = w_L, \qquad \tilde{\alpha} = \alpha^{\alpha}(1-\alpha)^{1-\alpha}, \tag{92}$$

$$c = \tilde{\theta} w_L(1-l_L)L + w_H(1-l_H)H, \qquad \tilde{\theta} = 1-\alpha+\theta\alpha, \tag{93}$$

$$R = \rho + g, \tag{94}$$

and

$$(R-g)k + w_L l_L L + w_H l_H H = c. \tag{95}$$

Combining these equilibrium conditions yields the following equation for low-skill labor supply:

$$(\tilde{\rho}+\tilde{\theta}+1)l_L^2 - \tilde{\theta} l_L + \frac{\tilde{\rho} l_H H + 2 l_H H - H}{\tilde{\alpha}\beta L} = 0, \tag{96}$$

where

$$\tilde{\rho} = \frac{\rho}{\dfrac{(\rho + g)^{(1-\eta)(1-\sigma)}}{1-n} - \rho - g}. \tag{97}$$

When $\rho < \tilde{\rho}$ and

$$\frac{H}{L} < \frac{\tilde{\alpha}\beta}{1 - 2\bar{l}_H},$$

Equation (96) admits a unique solution satisfying $l_L \in (0, 1)$. Hence, the balanced growth path is well defined. Along this path, the common growth rate of aggregate output, consumption, capital, and wages is

$$g = A_H \Delta,$$

while R, l_L, and the skill premium $w = w_H / w_L$ remain constant. This proves the existence and uniqueness of the balanced growth path stated in Proposition 1.

B.2 Proof of Proposition 2

This proof corresponds to Proposition 2 in the main text.

In the short run, all variables are treated as time independent. Combining the market-clearing conditions for low- and high-skill labor yields

$$L \int_M^N c \left(\frac{W_H}{\gamma_H(i)} \right)^{1-\sigma} W_H^{-1} \mathrm{d}i - H \int_I^M c \left(\frac{W_L}{\gamma_L(i)} \right)^{1-\sigma} W_L^{-1} \mathrm{d}i = 0. \tag{98}$$

Taking the total differential of Equation (98) gives

$$\begin{aligned}
& Lc\left(\frac{W_H}{\gamma_H(N)}\right)^{1-\sigma} W_H^{-1}\mathrm{d}N + Hc\left(\frac{W_L}{\gamma_L(I)}\right)^{1-\sigma} W_L^{-1}\mathrm{d}I \\
& \quad - \left[Hc\left(\frac{W_L}{\gamma_L(M)}\right)^{1-\sigma} W_L^{-1} + Lc\left(\frac{W_H}{\gamma_H(M)}\right)^{1-\sigma} W_H^{-1}\right]\mathrm{d}M \\
& \quad + L\int_M^N c\left(\frac{W_H}{\gamma_H(i)}\right)^{1-\sigma} (-\sigma) W_H^{-2}\mathrm{d}i\,\mathrm{d}W_H \\
& \quad - H\int_I^M c\left(\frac{W_L}{\gamma_L(i)}\right)^{1-\sigma} (-\sigma) W_L^{-2}\mathrm{d}i\,\mathrm{d}W_L = 0.
\end{aligned} \tag{99}$$

Equivalently,

$$\begin{aligned}
& Lc\left(\frac{W_H}{\gamma_H(N)}\right)^{1-\sigma} W_H^{-1}\mathrm{d}N + Hc\left(\frac{W_L}{\gamma_L(I)}\right)^{1-\sigma} W_L^{-1}\mathrm{d}I \\
& \quad - \left[Hc\left(\frac{W_L}{\gamma_L(M)}\right)^{1-\sigma} W_L^{-1} + Lc\left(\frac{W_H}{\gamma_H(M)}\right)^{1-\sigma} W_H^{-1}\right]\mathrm{d}M \\
& = \sigma L\int_M^N c\left(\frac{W_H}{\gamma_H(i)}\right)^{1-\sigma} W_H^{-1}\mathrm{d}i\,\left[\mathrm{d}\ln W_H - \mathrm{d}\ln W_L\right],
\end{aligned} \tag{100}$$

where Equation (98) has been used.

By the definition of the task-allocation threshold M, we have

$$\frac{W_H}{W_L} = \frac{\gamma_H(M)}{\gamma_L(M)}. \tag{101}$$

Under Assumption 1,

$$\frac{\mathrm{d}\ln(W_H/W_L)}{\mathrm{d}M} > 0. \tag{102}$$

To simplify notation, define

$$X_1 = Lc\left(\frac{W_H}{\gamma_H(N)}\right)^{1-\sigma} W_H^{-1} > 0, \tag{103}$$

$$X_2 = Hc\left(\frac{W_L}{\gamma_L(I)}\right)^{1-\sigma} W_L^{-1} > 0, \tag{104}$$

$$X_3 = Hc\left(\frac{W_L}{\gamma_L(M)}\right)^{1-\sigma} W_L^{-1} + Lc\left(\frac{W_H}{\gamma_H(M)}\right)^{1-\sigma} W_H^{-1} > 0, \tag{105}$$

and

$$X_4 = \sigma L \int_M^N c\left(\frac{W_H}{\gamma_H(i)}\right)^{1-\sigma} W_H^{-1} \mathrm{d}i \equiv X_0 > 0. \tag{106}$$

Using these definitions, Equation (100) implies

$$X_1 \mathrm{d}N + X_2 \mathrm{d}I - X_3 \mathrm{d}M = X_4 \mathrm{d}\ln\left(\frac{W_H}{W_L}\right). \tag{107}$$

Combining Equations (102) and (107), we obtain

$$\frac{\partial \ln(W_H/W_L)}{\partial N} = \frac{X_1}{X_3 + X_4/X_0} > 0, \qquad \frac{\partial \ln(W_H/W_L)}{\partial I} = \frac{X_2}{X_3 + X_4/X_0} > 0. \tag{108}$$

Therefore,

$$\frac{\partial w}{\partial N} > 0, \qquad \frac{\partial w}{\partial I} > 0.$$

Thus, when the capital stock is fixed in the short run, AI-induced task creation and automation both raise the skill premium. This proves Proposition 2.

B.3 Proof of Proposition 3

This proof corresponds to Proposition 3 in the main text.

In the long-run economy, after combining the equilibrium conditions, the final equation governing low-skill labor supply is

$$(\tilde{\rho} + \tilde{\theta} + 1) l_L^2 - \tilde{\theta} l_L + \frac{\tilde{\rho} l_H H + 2 l_H H - H}{\tilde{\alpha}\beta L} = 0, \tag{109}$$

where

$$\tilde{\rho} = \frac{\rho}{\dfrac{(\rho + g)^{(1-\eta)(1-\sigma)}}{1-n} - \rho - g}. \tag{110}$$

Taking the total differential of Equation (109) with respect to l_L and $\tilde{\rho}$ gives

$$\mathrm{d}l_L + \left[l_L^2 + \frac{l_L H}{\tilde{\alpha}\beta L} \right] \mathrm{d}\tilde{\rho} = 0. \tag{111}$$

Since

$$\frac{\mathrm{d}\tilde{\rho}}{\mathrm{d}n} < 0, \tag{112}$$

it follows from Equation (111) that

$$\frac{\mathrm{d}l_L}{\mathrm{d}n} > 0. \tag{113}$$

Moreover, the long-run skill-premium condition is

$$\tilde{\alpha}\beta l_L w = 1. \tag{114}$$

Therefore,

$$\frac{\mathrm{d}w}{\mathrm{d}n} < 0. \tag{115}$$

By definition, $n = N - I$. Hence,

$$\frac{\partial n}{\partial N} > 0, \qquad \frac{\partial n}{\partial I} < 0.$$

Combining these relations with Equation (115) yields

$$\frac{\partial w}{\partial N} < 0, \qquad \frac{\partial w}{\partial I} > 0.$$

Thus, in the long run, the creation effect reduces the skill premium, whereas the substitution effect increases it. This proves Proposition 3.

B.4 Proof of Proposition 4

This proof corresponds to Proposition 4 in the main text.

We use a local linearization argument to study the stability of the balanced growth path with endogenous technological change. Consider the case in which $S < \bar{S}$, so that the equilibrium growth rate g converges to zero in the neighborhood of the steady state. For any initial AI technology level $n(0)$ and capital stock $k(0)$, an equilibrium with endogenous technological change is characterized by the dynamic system in $\{n, k, c, v\}$.

First, the evolution of AI technology is determined by the allocation of scientists across research directions:

$$\dot{n}(t) = \kappa_N S - [\phi_I(n)\kappa_I + \kappa_N \phi_N(n)]\, G(v)S. \tag{116}$$

Second, household consumption satisfies the Euler equation

$$\frac{\dot{c}(t)}{c(t)} = \frac{1}{\theta}\left[R^E(n(t), k(t)) - \delta - \rho\right] - O(g). \tag{117}$$

Third, the capital stock evolves according to the aggregate resource constraint:

$$\dot{k}(t) = f^E(n(t), k(t)) - c(t) - \delta k(t). \tag{118}$$

Finally, the normalized value differential $v(t)$ satisfies

$$\begin{aligned}\dot{v}(t) = &\left[\kappa_I \phi_I'(n) v_I(n) + \kappa_N \phi_N'(n) v_N(n)\right] \left[\kappa_N S - (\phi_I(n)\kappa_I + \phi_N(n)\kappa_N)\, G(v)S\right] \\ &+ \rho v + \kappa_N \phi_N(n)\pi_N(n) - \kappa_I \phi_I(n)\pi_I(n) + O(g).\end{aligned} \tag{119}$$

Let n^α, $v^\alpha = 0$, c^α, and k^α denote the values of the corresponding variables at the interior balanced growth path. The dynamic system can be locally approximated around the equilibrium as follows:

$$\dot{n} = P_n(n(t) - n^\alpha) + P_v v(t), \tag{120}$$

$$\dot{v} = Q_n(n(t) - n^\alpha) + Q_v v(t) + Q_k(k(t) - k^\alpha), \tag{121}$$

$$\dot{c} = \frac{c^\alpha}{\theta} R_n^E(n(t) - n^\alpha) + \frac{c^\alpha}{\theta} R_k^E(k(t) - k^\alpha), \tag{122}$$

$$\dot{k} = f_n^E(n(t) - n^\alpha) + (f_k^E - \delta)(k(t) - k^\alpha) - (c(t) - c^\alpha). \tag{123}$$

Here P_n, P_v, Q_n, and Q_v are defined consistently with the dynamic system above, while R_n^E, R_k^E, f_n^E, and f_k^E denote the derivatives of R^E and f^E with respect to n and k, evaluated at the balanced growth path. Moreover,

$$Q_k = \kappa_N \frac{\partial \pi_N(n,k)}{\partial k} - \kappa_I \frac{\partial \left[\phi_I(n)\pi_I(n,k)\right]}{\partial k}\bigg|_{k=k^\alpha}. \tag{124}$$

The corresponding characteristic equation is

$$F(\lambda) = \begin{vmatrix} P_n - \lambda & P_v & 0 & 0 \\ Q_n & Q_v - \lambda & 0 & Q_k \\ \frac{c^\alpha}{\theta} R_n^E & 0 & -\lambda & \frac{c^\alpha}{\theta} R_k^E \\ f_n^E & 0 & -1 & f_k^E - \delta - \lambda \end{vmatrix}. \tag{125}$$

Expanding the determinant gives

$$\begin{aligned} F(\lambda) =& \lambda^4 + (-P_n - Q_v - f_k^E + \delta)\lambda^3 \\ &+ \left[P_n Q_v + (P_n + Q_v)(f_k^E - \delta) + \frac{c^\alpha}{\theta} R_k^E - P_v Q_n\right]\lambda^2 \\ &+ \left[P_n Q_v(\delta - f_k^E) - (P_n + Q_v)\frac{c^\alpha}{\theta} R_k^E + P_v\left(Q_n f_k^E - Q_n \delta - Q_k f_n^E\right)\right]\lambda \\ &+ \left[P_n Q_v \frac{c^\alpha}{\theta} R_k^E + P_v \frac{c^\alpha}{\theta}\left(R_n^E Q_k - R_k^E Q_n\right)\right]. \end{aligned} \tag{126}$$

By Vieta's formulas,

$$\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = P_n + Q_v + f_k^E - \delta, \tag{127}$$

$$\sum_{i<j} \lambda_i \lambda_j = P_n Q_v + (P_n + Q_v)(f_k^E - \delta) + \frac{c^\alpha}{\theta} R_k^E - P_v Q_n, \tag{128}$$

$$\sum_{i<j<k} \lambda_i \lambda_j \lambda_k = P_n Q_v (f_k^E - \delta) + (P_n + Q_v)\frac{c^\alpha}{\theta} R_k^E + P_v \left[Q_k f_n^E - Q_n (f_k^E - \delta) \right], \tag{129}$$

and

$$\lambda_1 \lambda_2 \lambda_3 \lambda_4 = P_n Q_v \frac{c^\alpha}{\theta} R_k^E + P_v \frac{c^\alpha}{\theta} \left(R_n^E Q_k - R_k^E Q_n \right). \tag{130}$$

We next show that, among the four characteristic roots, exactly two have positive real parts and two have negative real parts. First, we establish that

$$\lambda_1 \lambda_2 \lambda_3 \lambda_4 > 0.$$

The market-clearing conditions imply $R_k^E < 0$. Moreover, the preceding stability conditions imply $P_n < 0$ and $Q_v > 0$. Hence,

$$P_n Q_v \frac{c^\alpha}{\theta} R_k^E > 0.$$

Since $P_v < 0$, it remains sufficient to show that

$$R_n^E Q_k - R_k^E Q_n < 0.$$

Equivalently,

$$Q_n - Q_k \frac{R_n^E}{R_k^E} < 0.$$

When $S < \bar{S}$, this condition is equivalent to the derivative with respect to n of

$$\kappa_N \phi_N(n) v_N(n, k(n)) - \kappa_I \phi_I(n) v_I(n, k(n))$$

being negative at $n = n^\alpha$. Geometrically, this means that the curve

$$\kappa_I \phi_I(n) v_I(n, k(n))$$

crosses the curve

$$\kappa_N \phi_N(n) v_N(n, k(n))$$

from below at the equilibrium point.

Therefore, the four characteristic roots can have only one of the following three sign configurations: all four have positive real parts, all four have negative real parts, or exactly two have positive real parts and two have negative real parts.

We now rule out the case in which all four roots have positive real parts. Let $\hat{\rho}$ denote the value for which

$$P_n + Q_v = 0.$$

Then, when $\rho > \hat{\rho}$, we have $P_n + Q_v > 0$. If $f_k^E - \delta < 0$, Equation (128) implies that the sum of all pairwise products cannot be positive. If $f_k^E - \delta > 0$, the sum of all triple products in Equation (129) cannot be positive. In either case, the configuration in which all four roots have positive real parts is ruled out. This conclusion also uses the fact that

$$Q_k f_n^E - Q_n(f_k^E - \delta) > 0,$$

which follows from the negative derivative condition at $n = n^\alpha$ stated above.

Finally, we rule out the case in which all four roots have negative real parts. If $f_k^E - \delta < 0$, then the sum of all pairwise products in Equation (128) cannot be positive. If $f_k^E - \delta > 0$, then

$$\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 > 0,$$

which contradicts the possibility that all four roots have negative real parts. Hence, the configuration in which all four roots have negative real parts is also impossible.

It follows that exactly two characteristic roots have positive real parts and two have negative real parts. Since the system contains two state variables, n and k, and two jump variables, c and v, this saddle-path structure implies local asymptotic stability of the balanced growth path with endogenous technological change. This proves Proposition 4.

Appendix C Derivation of Value Functions

This appendix derives the optimality conditions and value functions for AI research firms. The derivation corresponds to the profit functions in Equations (33) and (34) and the value functions in Equations (36) and (37) in the main text.

C.1 Profits and Value Functions of AI Research Firms

This subsection derives the profits and value functions associated with job-replacing and job-creating technological products. The derivation links back to the AI research sector described in Section 4.1.

First, using the demand function for the technological product $q(i,t)$ derived in Section A.2, together with the profit earned on each unit of $q(i,t)$, namely $(1-\mu)\psi$, the total profit from a technological product can be obtained as follows:

$$q(i,t) = \eta\psi^{-1}y(i,t)c(\cdot), \tag{131}$$

and

$$y(i,t) = Y(t)c(\cdot)^{-\sigma}. \tag{132}$$

Therefore, the profit flow generated by technological product $q(i,t)$ is

$$\begin{aligned}\pi(i,t) &= \eta\psi^{-1}Y(t)c(\cdot)^{1-\sigma}(1-\mu)\psi \\ &= (1-\mu)\eta Y(t)c(\cdot)^{1-\sigma}.\end{aligned} \tag{133}$$

Accordingly, the profit flow associated with a job-replacing technological product is

$$\pi_I(i,t) = (1-\mu)\eta Y(t)c(R(t))^{1-\sigma}. \tag{134}$$

Similarly, the profit flow associated with a job-creating technological product is

$$\begin{aligned}\pi_N(i,t) = (1-\mu)\eta Y(t)\Bigg[& c\left(\frac{W_L(t)}{\gamma_L(i,t)}\right)^{1-\sigma}\frac{\operatorname{sgn}(M(t)-i)+1}{2} \\ & + c\left(\frac{W_H(t)}{\gamma_H(i,t)}\right)^{1-\sigma}\frac{\operatorname{sgn}(i-M(t))+1}{2}\Bigg].\end{aligned} \tag{135}$$

These expressions correspond to Equations (33) and (34) in the main text.

The value functions must satisfy the Bellman equations

$$r(t)V_I(i,t) - \dot{V}_I(i,t) = \pi_I(i,t), \tag{136}$$

and

$$r(t)V_N(i,t) - \dot{V}_N(i,t) = \pi_N(i,t). \tag{137}$$

Equivalently, Equation (136) can be written as

$$\frac{\partial V_I(i,t)}{\partial t} - r(t)V_I(i,t) = -\pi_I(i,t). \tag{138}$$

Solving this linear differential equation under the terminal condition

$$V_I(i,T^I(i)) = 0 \tag{139}$$

yields

$$V_I(i,t) = \int_t^{T^I(i)} e^{-\int_t^\tau r(s)\mathrm{d}s}\pi_I(i,\tau)\mathrm{d}\tau. \tag{140}$$

Since the effective discount rate is given by

$$r(s) = R(s) - \delta,$$

we obtain

$$V_I(i,t) = \int_t^{T^I(i)} e^{-\int_t^\tau [R(s)-\delta]\mathrm{d}s} \pi_I(i,\tau)\mathrm{d}\tau. \tag{141}$$

Substituting Equation (134) into Equation (141) gives

$$V_I(I(t),t) = (1-\mu)\eta \int_t^{T^I(I(t))} e^{-\int_t^\tau [R(s)-\delta]\mathrm{d}s} Y(\tau)c(R(\tau))^{1-\sigma}\mathrm{d}\tau. \tag{142}$$

Similarly, the Bellman equation for job-creating technological products can be written as

$$\frac{\partial V_N(i,t)}{\partial t} - r(t)V_N(i,t) = -\pi_N(i,t). \tag{143}$$

Under the terminal condition

$$V_N(i,T^N(i)) = 0, \tag{144}$$

the solution is

$$V_N(i,t) = \int_t^{T^N(i)} e^{-\int_t^\tau [R(s)-\delta]\mathrm{d}s} \pi_N(i,\tau)\mathrm{d}\tau. \tag{145}$$

For a newly created frontier intermediate good $i = N(t)$, production is initially carried out by high-skill labor. Once low-skill workers become sufficiently familiar with the task, production shifts from high-skill labor to low-skill labor at time $T^M(N(t))$. Therefore, the value of a job-creating technological product can be decomposed into two components:

$$\begin{aligned} V_N(N(t),t) =& (1-\mu)\eta \int_t^{T^M(N(t))} e^{-\int_t^\tau [R(s)-\delta]\mathrm{d}s} Y(\tau)c\left(\frac{W_H(\tau)}{\gamma_H(N(t))}\right)^{1-\sigma} \mathrm{d}\tau \\ &+ (1-\mu)\eta \int_{T^M(N(t))}^{T^N(N(t))} e^{-\int_t^\tau [R(s)-\delta]\mathrm{d}s} Y(\tau)c\left(\frac{W_L(\tau)}{\gamma_L(N(t),\tau)}\right)^{1-\sigma} \mathrm{d}\tau. \end{aligned} \tag{146}$$

The first term captures the profit flow obtained while the newly created intermediate good is produced by high-skill labor. The second term captures the profit flow obtained after low-skill workers acquire sufficient task-specific familiarity and production shifts to low-skill labor. Equations Equations (142) and (146) correspond to Equations (36) and (37) in the main text.

C.2 Derivation of Normalized Value Functions

This subsection derives the normalized value functions used in the endogenous technological progress model. The derivation links back to the normalized values defined in Equation (43).

Consider first the value of a job-replacing technological product:

$$V_I(I(t),t) = (1-\mu)\eta \int_t^{T^I(I(t))} e^{-\int_t^\tau [R(s)-\delta]\mathrm{d}s} Y(\tau)c(R(\tau))^{1-\sigma}\mathrm{d}\tau. \tag{147}$$

Along the balanced growth path, the lifetime of a technological product is constant. Specifically,

$$T^I(I(t)) - t = \frac{1-n}{\Delta}, \tag{148}$$

where

$$\Delta = \frac{\kappa_I\phi_I(n^\alpha)\kappa_N\phi_N(n^\alpha)}{\kappa_I\phi_I(n^\alpha)+\kappa_N\phi_N(n^\alpha)}S \tag{149}$$

denotes the equilibrium research speed.

Let $\tau = t+s$. Dividing Equation (147) by $Y(t)$ gives the normalized value of a job-replacing technological product:

$$v_I(n) = (1-\mu)\eta \int_0^{\frac{1-n}{\Delta}} e^{-\int_t^{t+s}[R(u)-\delta-g(u)]\mathrm{d}u} c(R(t+s))^{1-\sigma}\mathrm{d}s, \tag{150}$$

where $g(u)$ denotes the growth rate of aggregate output, so that

$$Y(t+s) = Y(t)\exp\left(\int_t^{t+s} g(u)\mathrm{d}u\right).$$

The same method can be applied to the value of a job-creating technological product. Along the balanced growth path, a newly created intermediate good is first produced by high-skill labor and then, after low-skill workers accumulate sufficient task-specific familiarity, by low-skill labor. The normalized value of a job-creating technological product is therefore

$$\begin{aligned} v_N(n) =& (1-\mu)\eta \int_0^{\frac{n-m}{\Delta}} e^{-\int_t^{t+s}[R(u)-\delta-g(u)]\mathrm{d}u} c\left(\frac{w_H(n)}{\gamma_H(n)} \exp\left\{\int_t^{t+s} g(u)\mathrm{d}u\right\}\right)^{1-\sigma} \mathrm{d}s \\ &+ (1-\mu)\eta \int_{\frac{n-m}{\Delta}}^{\frac{n}{\Delta}} e^{-\int_t^{t+s}[R(u)-\delta-g(u)]\mathrm{d}u} c\left(\frac{w_L(n)}{\gamma_L(n)} \exp\left\{\int_t^{t+s} g(u)\mathrm{d}u\right\}\right)^{1-\sigma} \mathrm{d}s. \end{aligned} \tag{151}$$

The first integral in Equation (151) corresponds to the period during which the newly created intermediate good is high-skill-labor intensive. The second integral corresponds to the period after learning by doing allows low-skill workers to produce the same intermediate good. Equations (150) and (151) provide the normalized value functions used in the endogenous technological progress model.